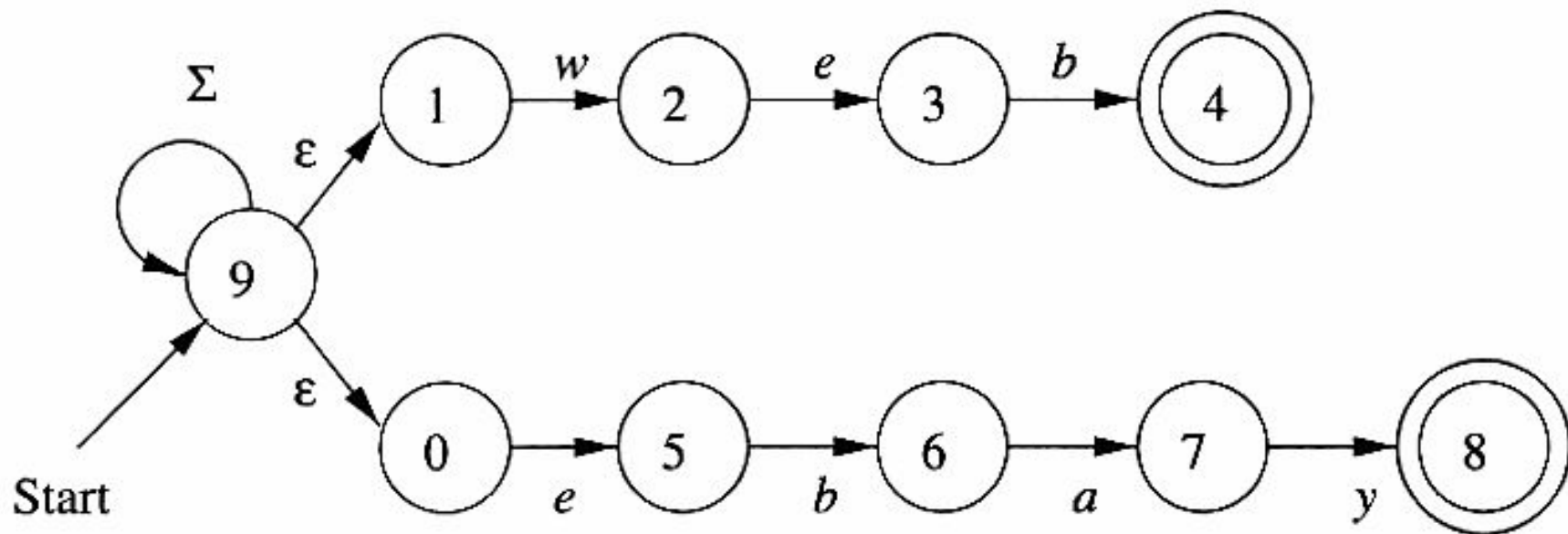
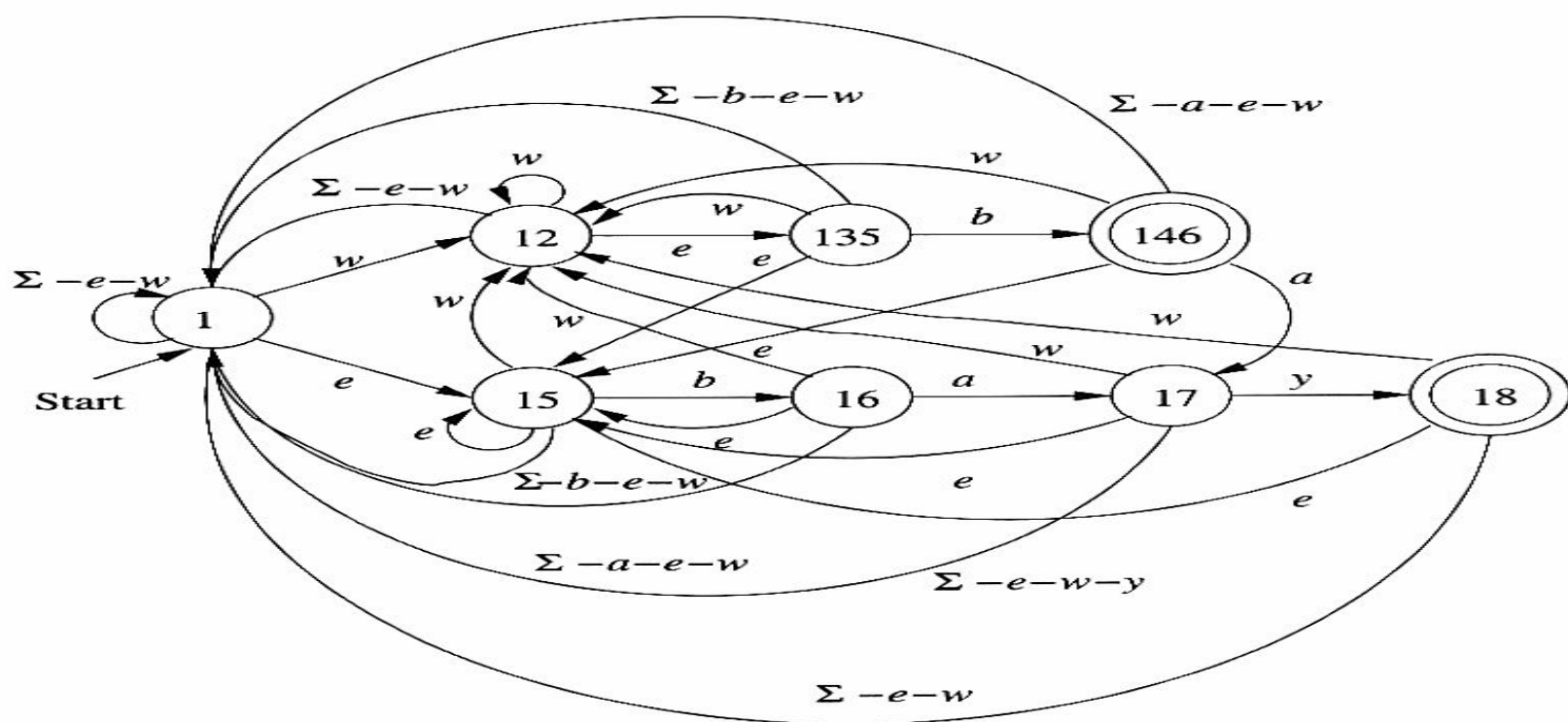
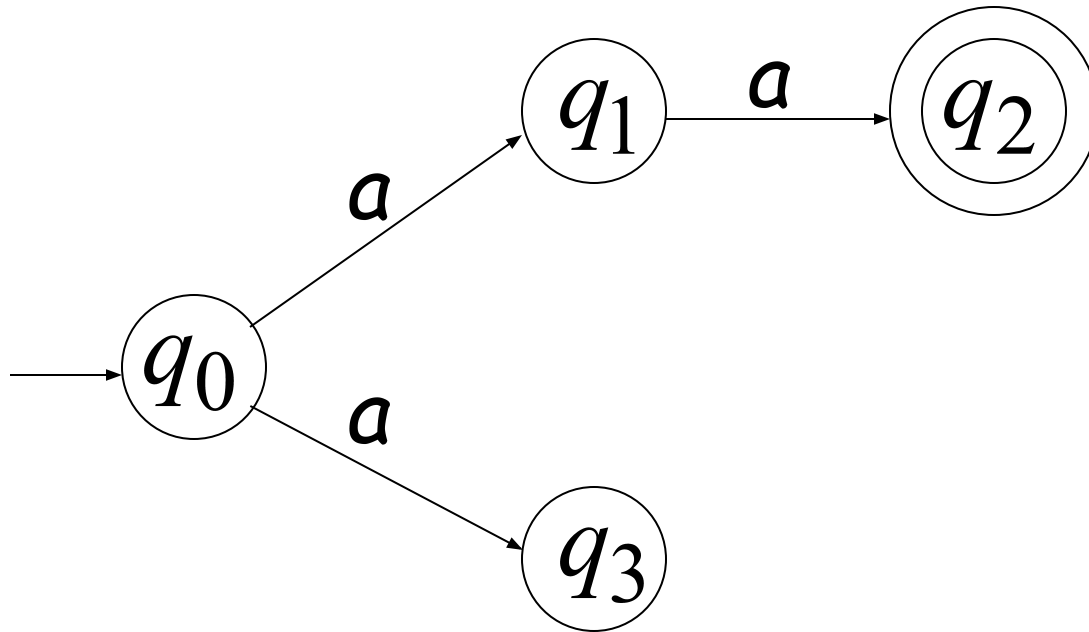




Non-Deterministic Finite Automata

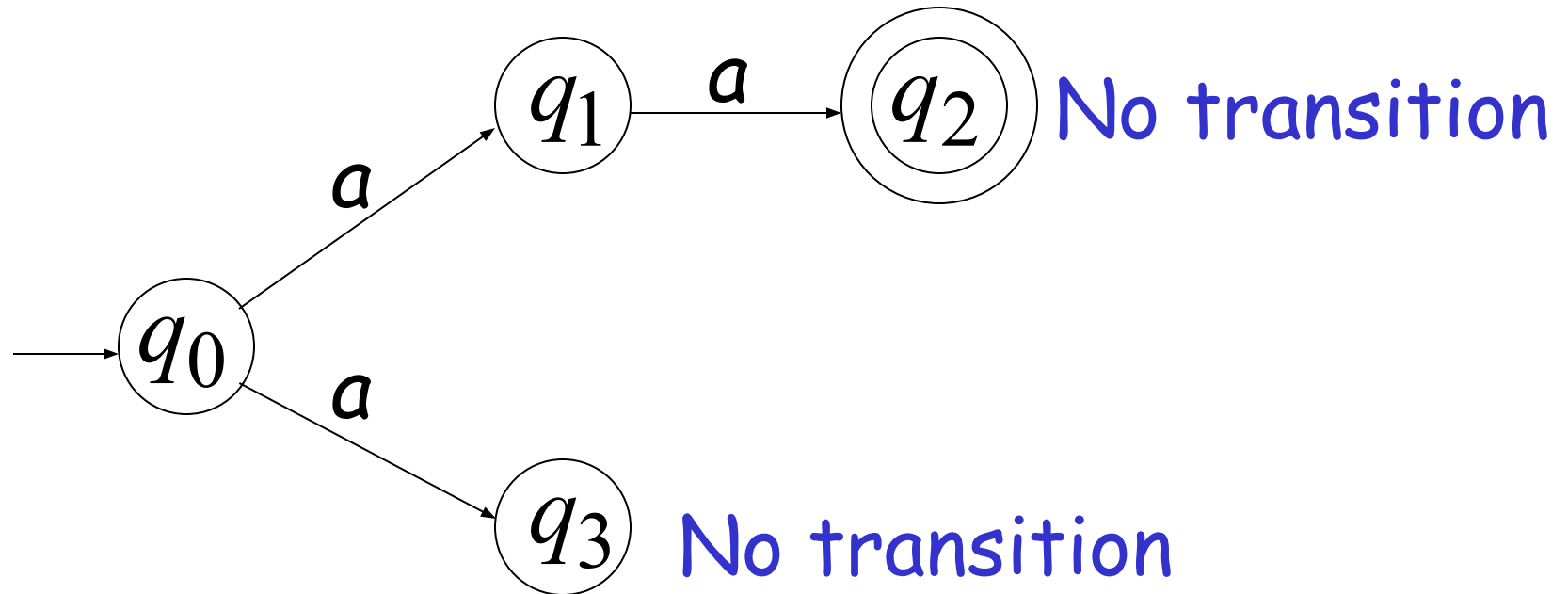
Nondeterministic Finite Acceptor (NFA)

Alphabet set = $\{a\}$



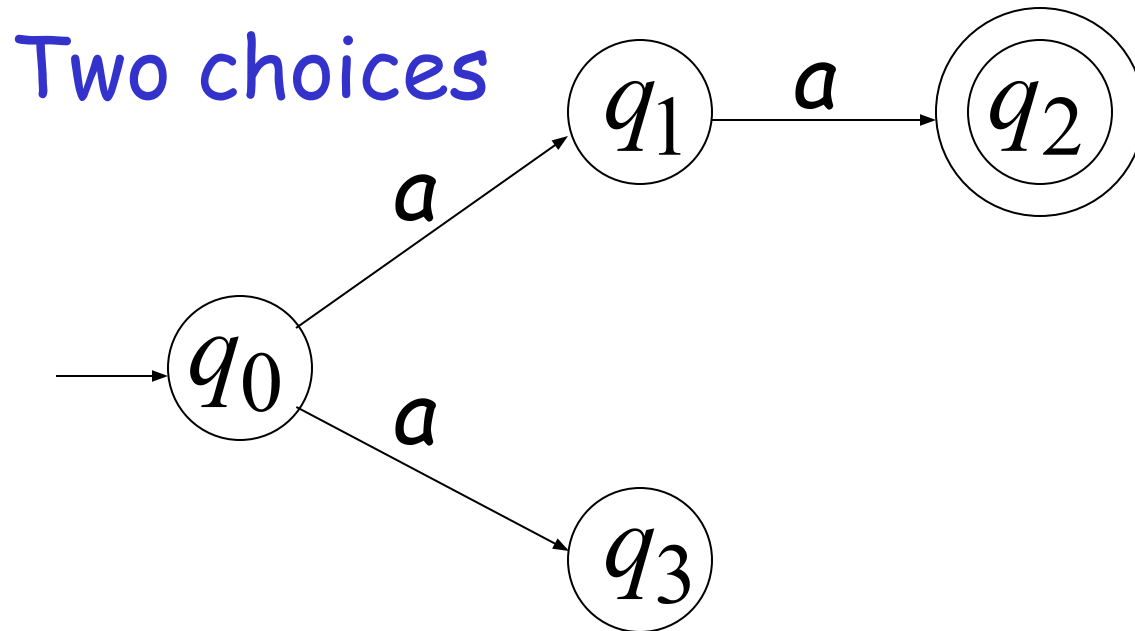
Nondeterministic Finite Acceptor (NFA)

Alphabet set = $\{a\}$

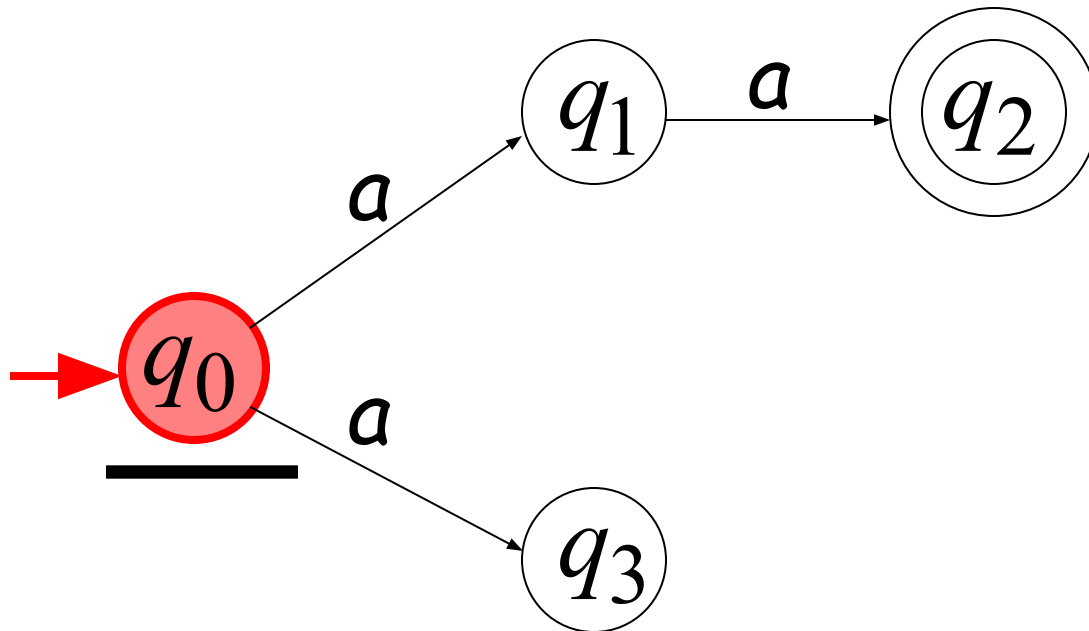
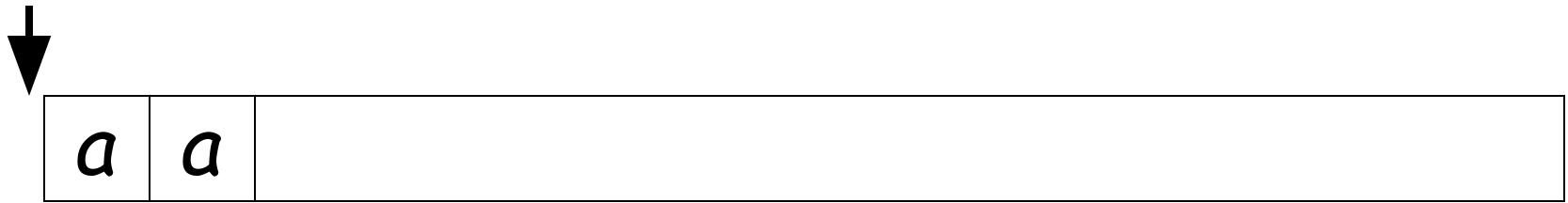


Nondeterministic Finite Acceptor (NFA)

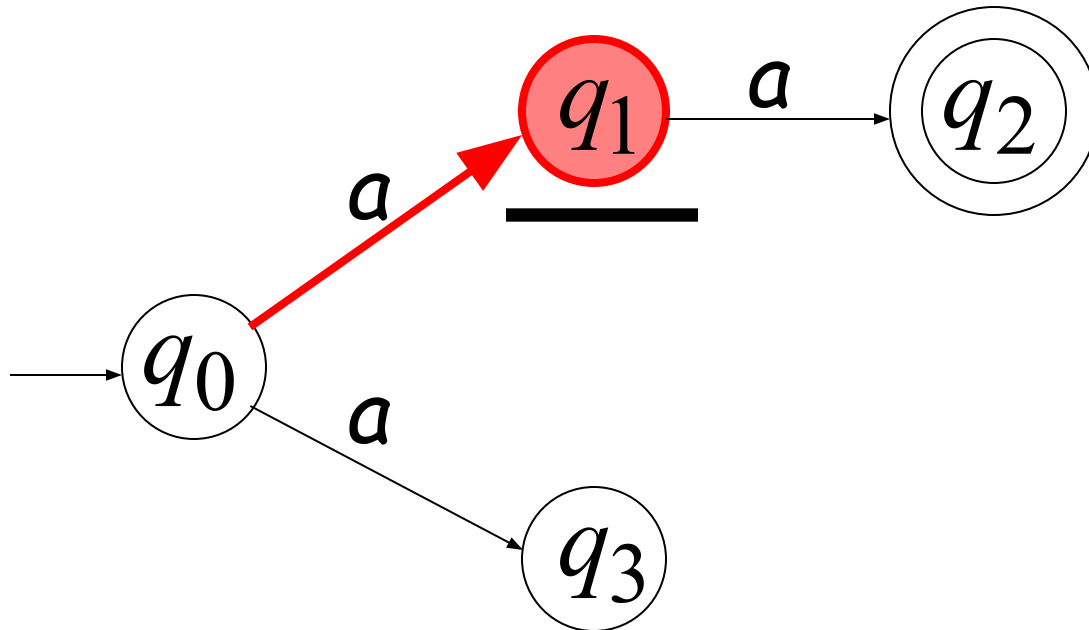
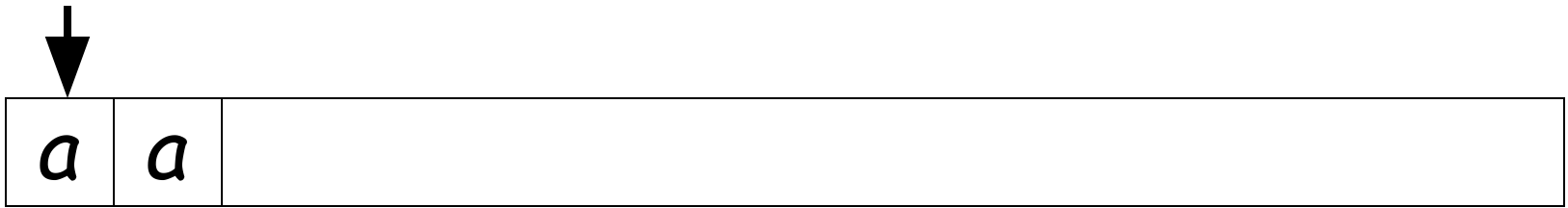
Alphabet set = $\{a\}$



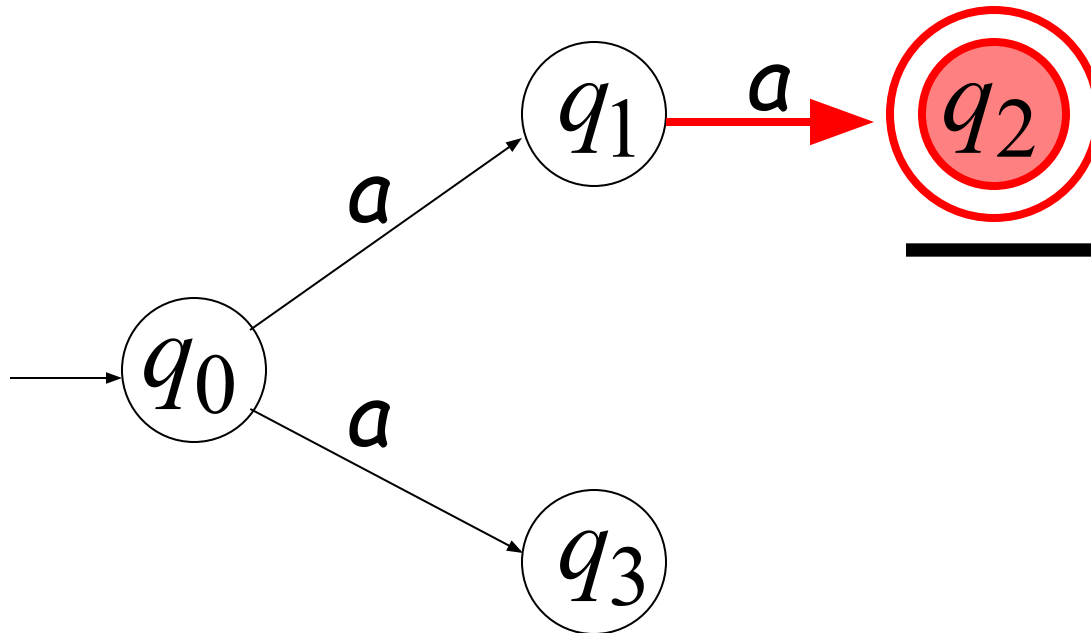
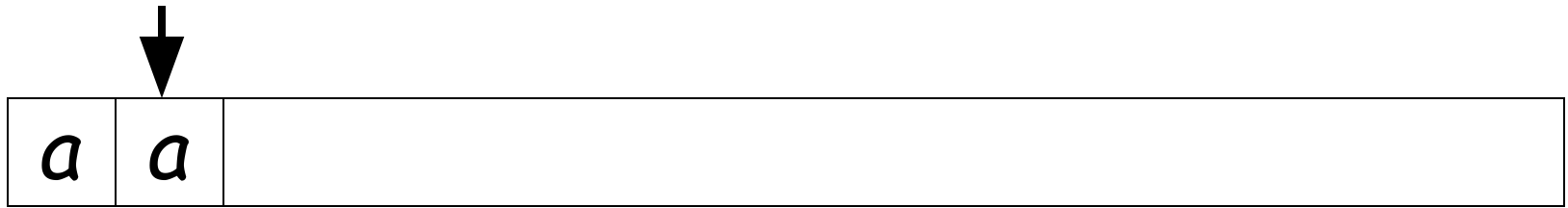
First Choice



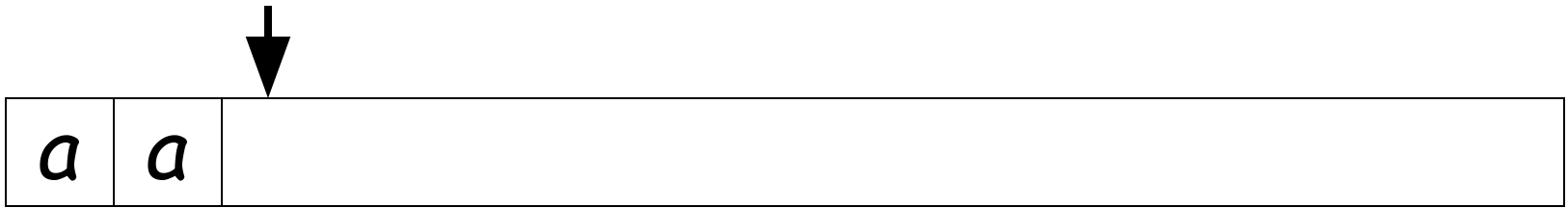
First Choice



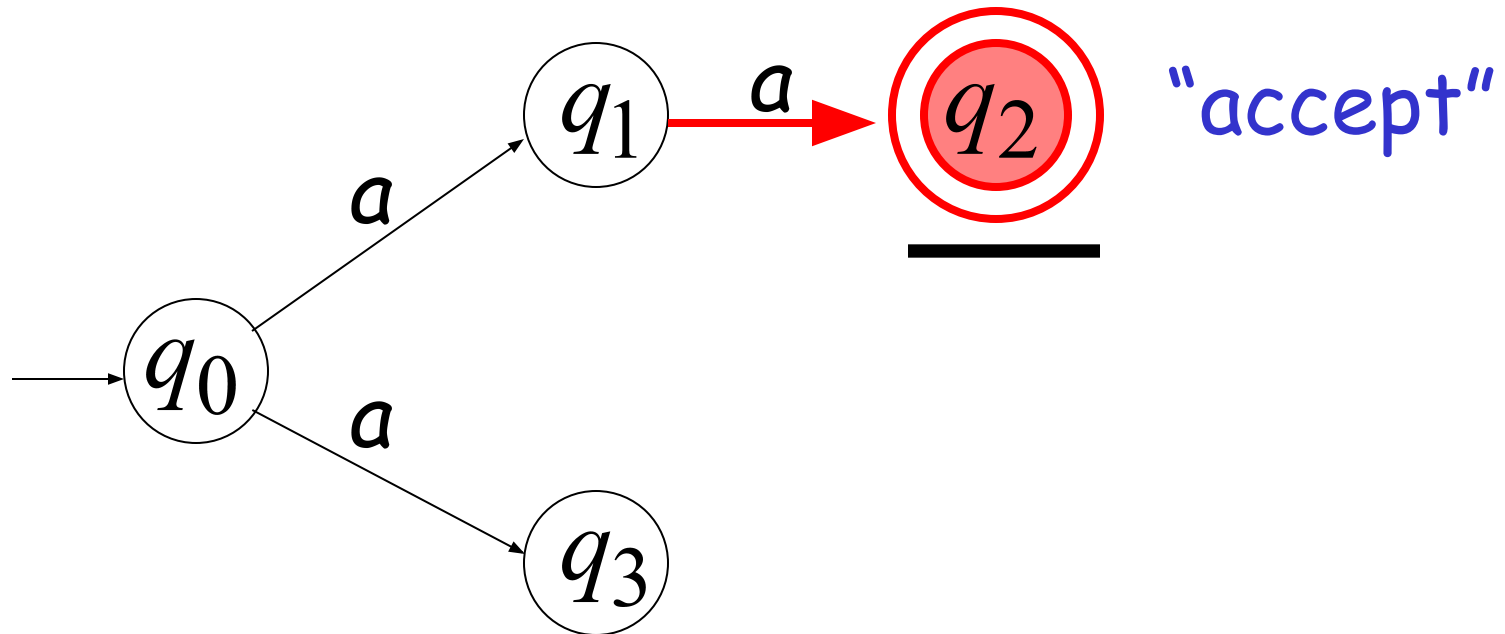
First Choice



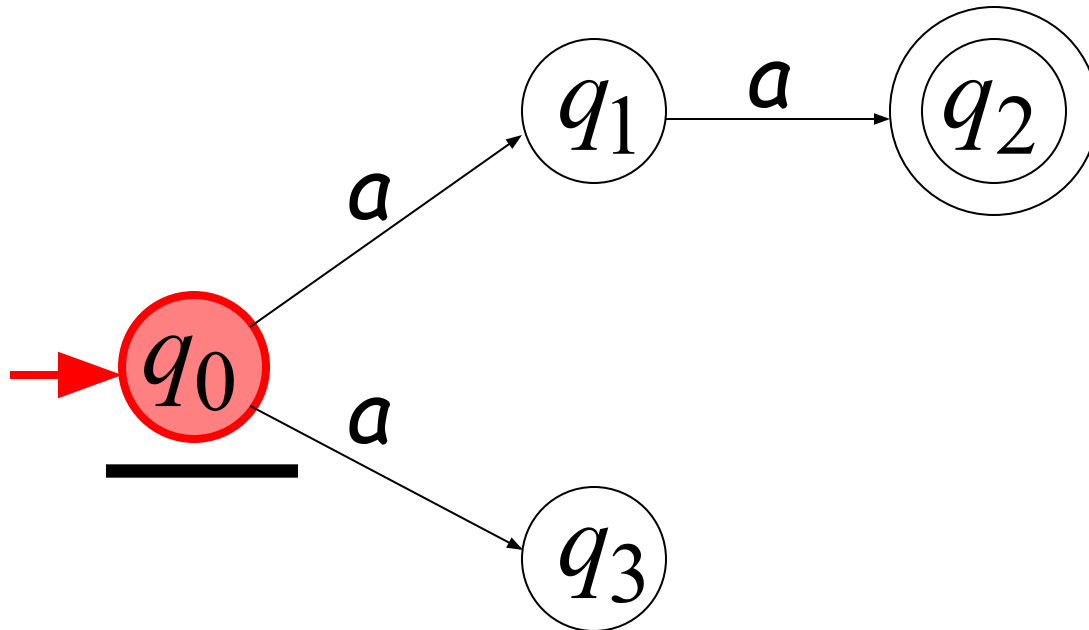
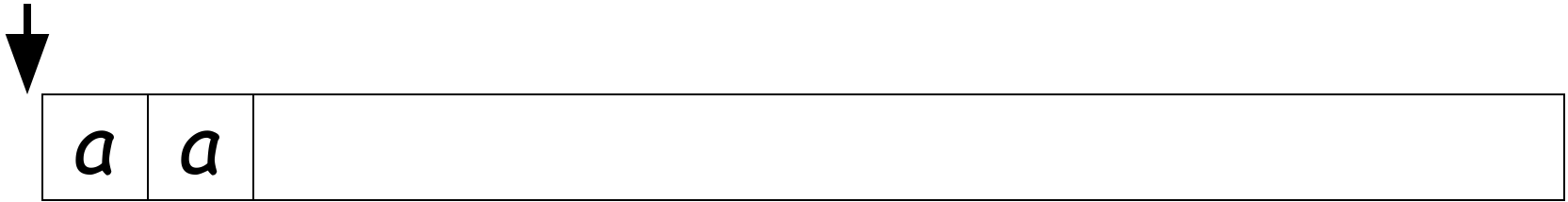
First Choice



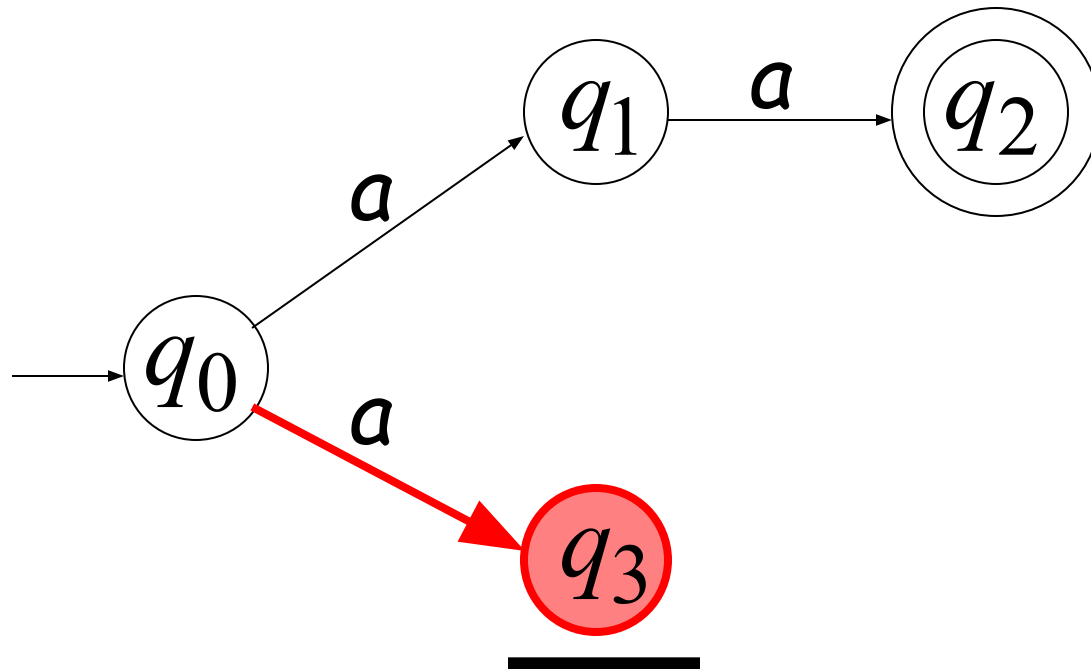
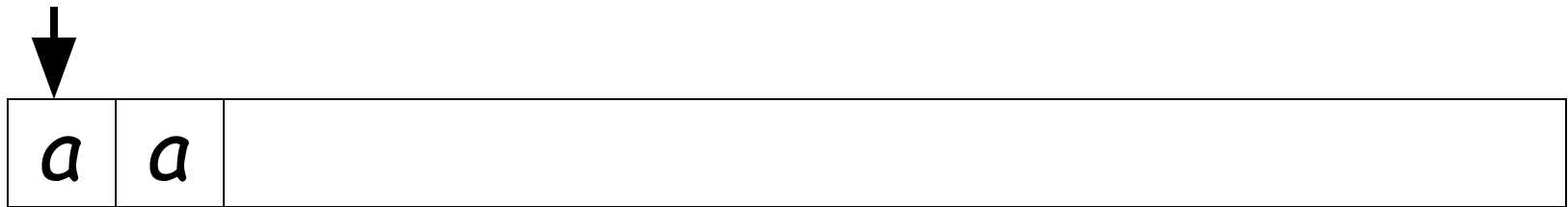
All input is consumed



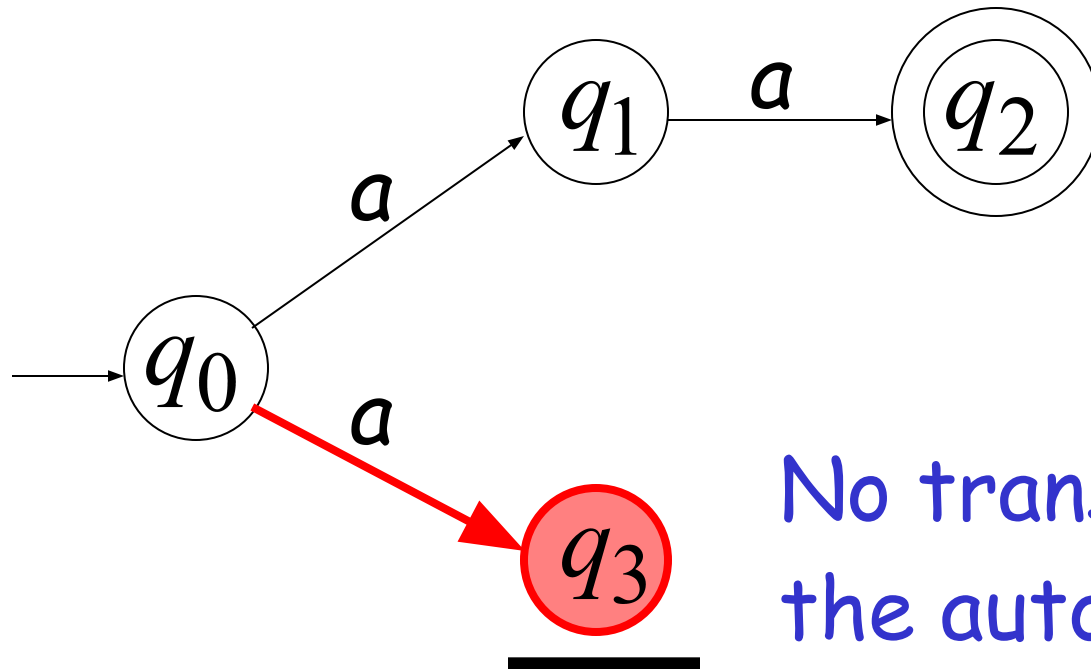
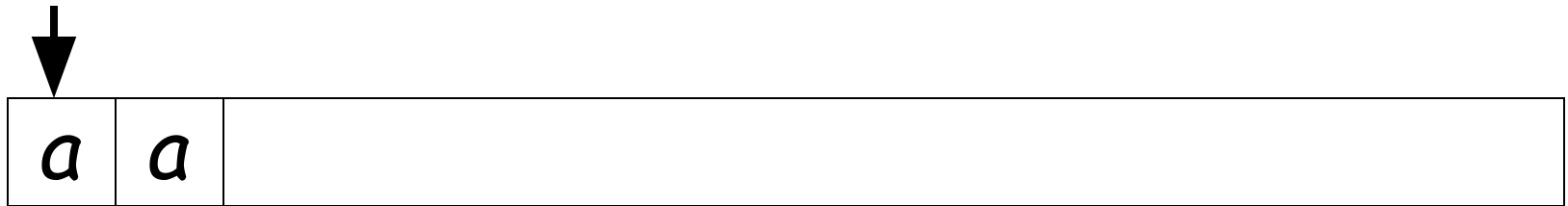
Second Choice



Second Choice

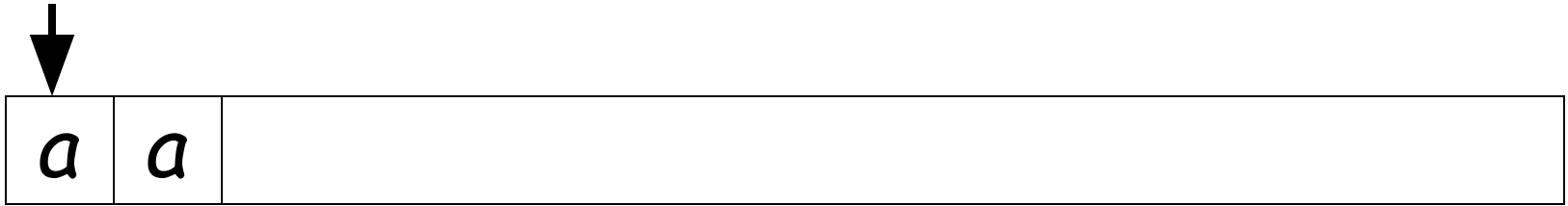


Second Choice

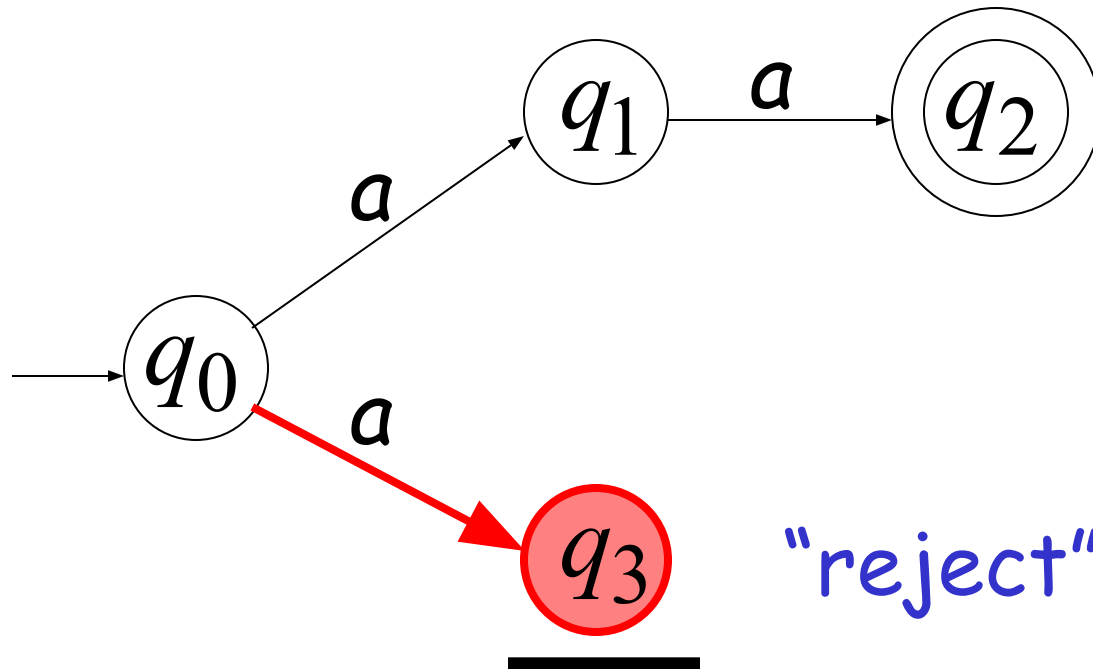


No transition:
the automaton hangs

Second Choice



Input cannot be consumed



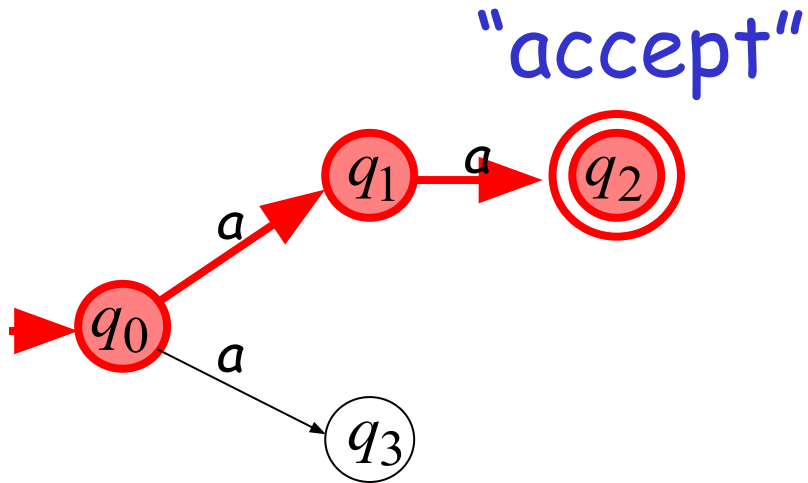
An NFA accepts a string:
when there is a computation of the NFA
that accepts the string

AND

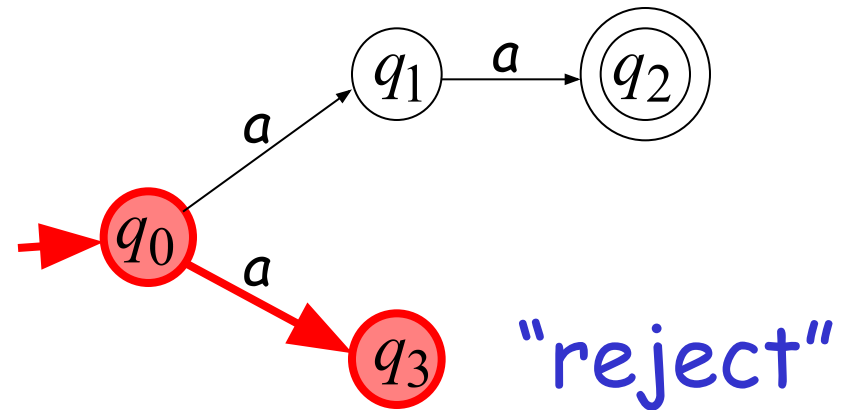
all the input is consumed and the automaton
is in a final state

Example

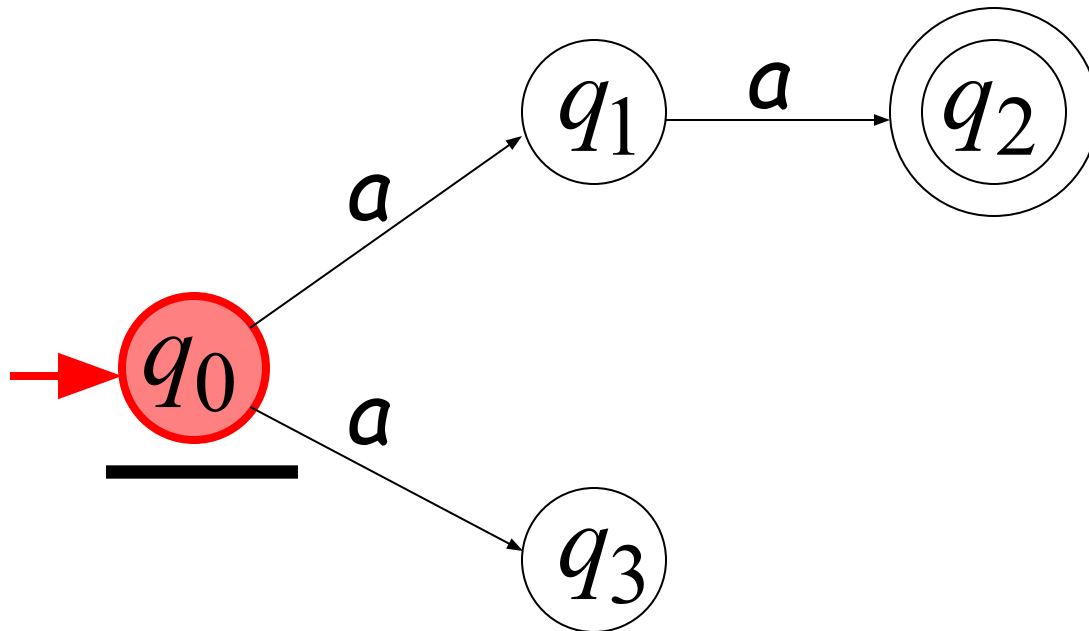
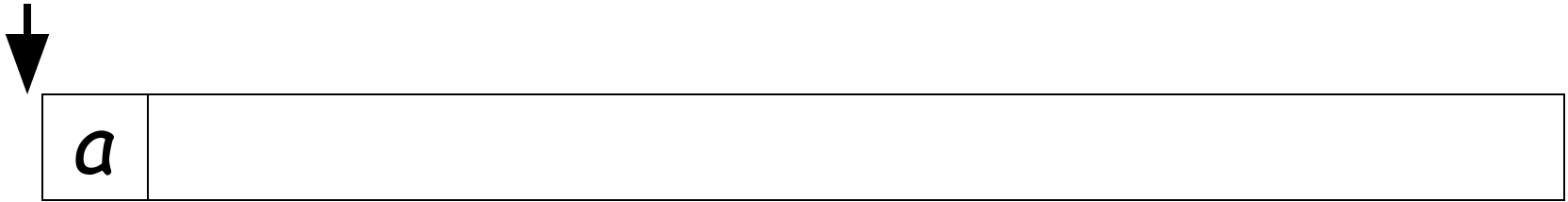
aa is accepted by the NFA:



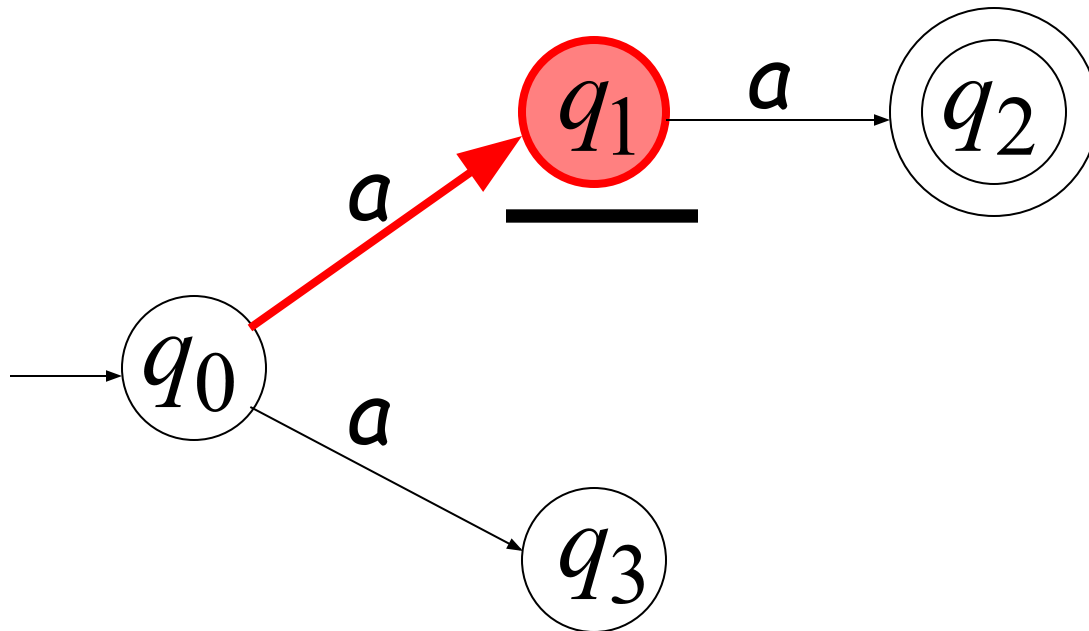
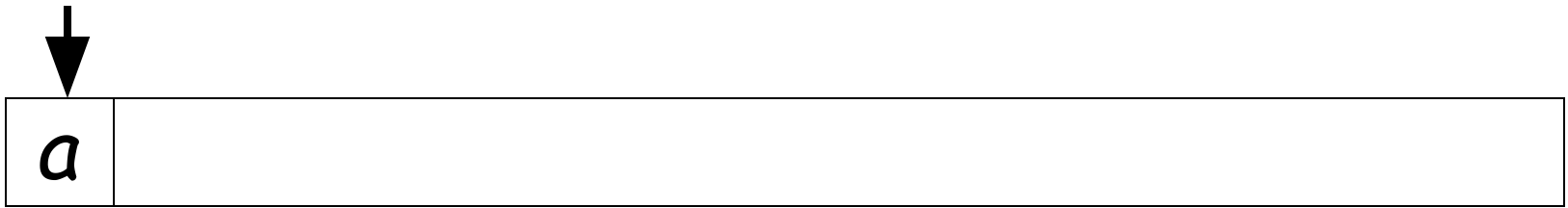
because this
computation
accepts aa



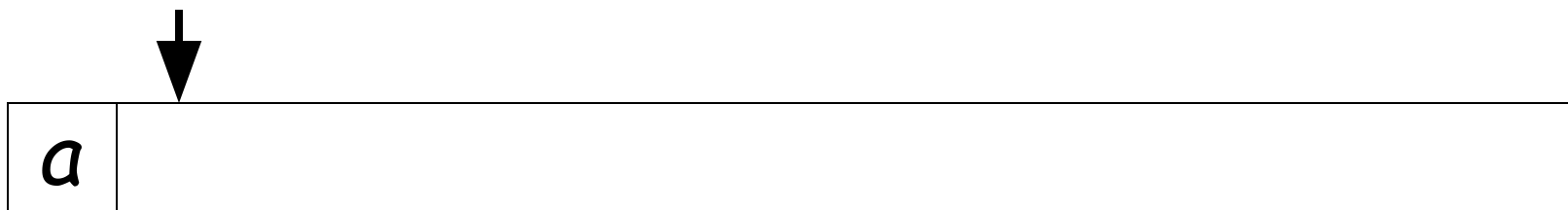
Rejection example



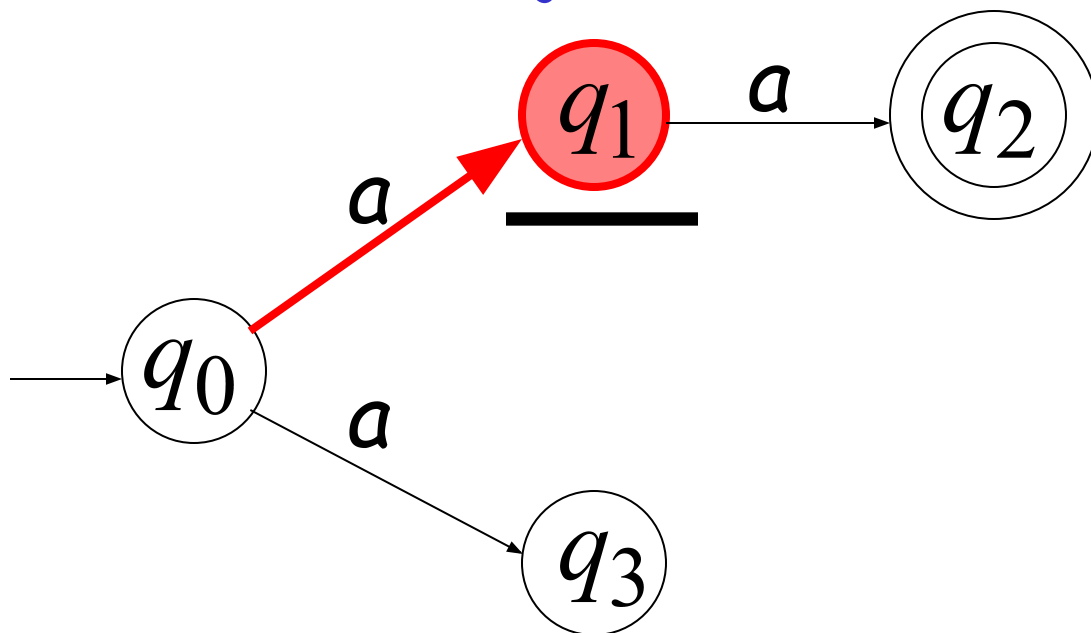
First Choice



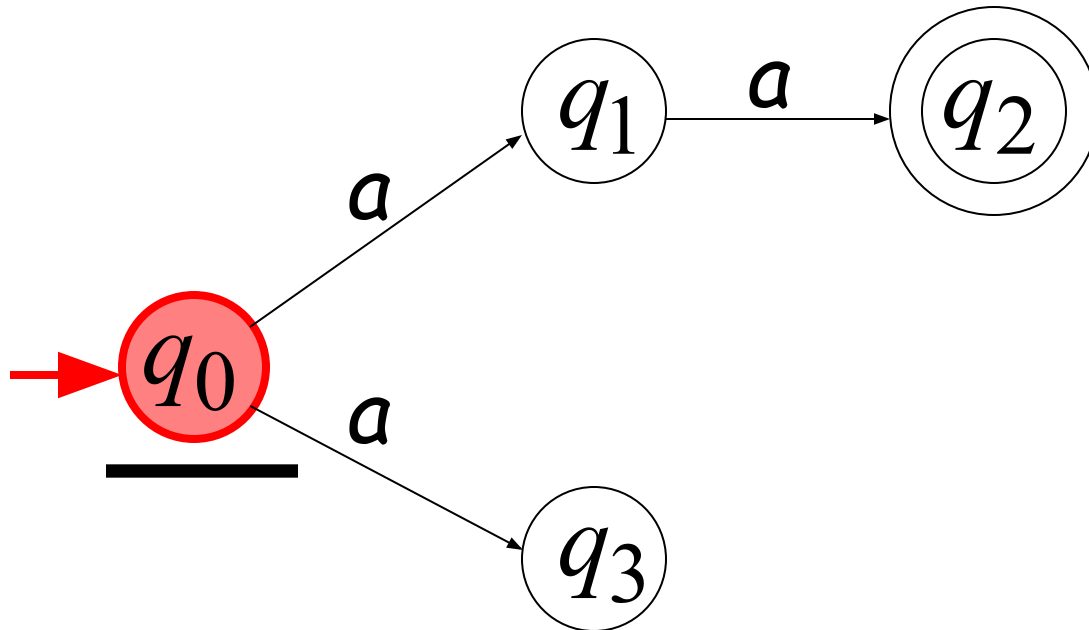
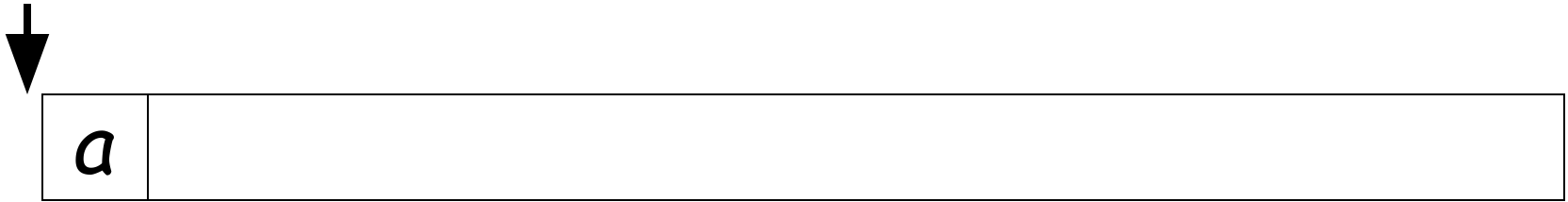
First Choice



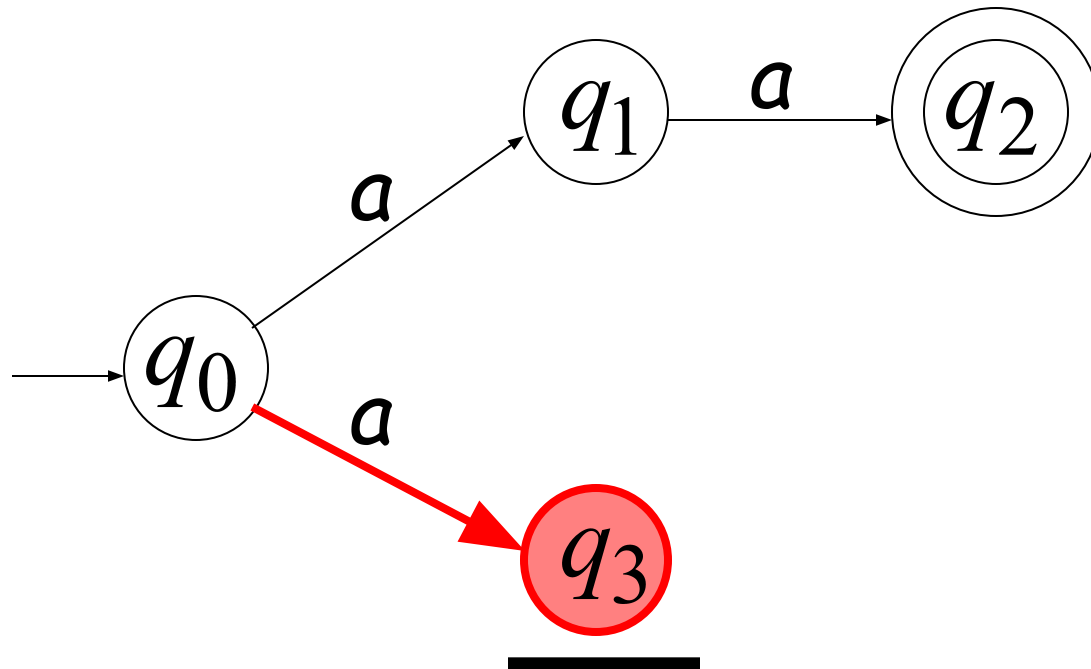
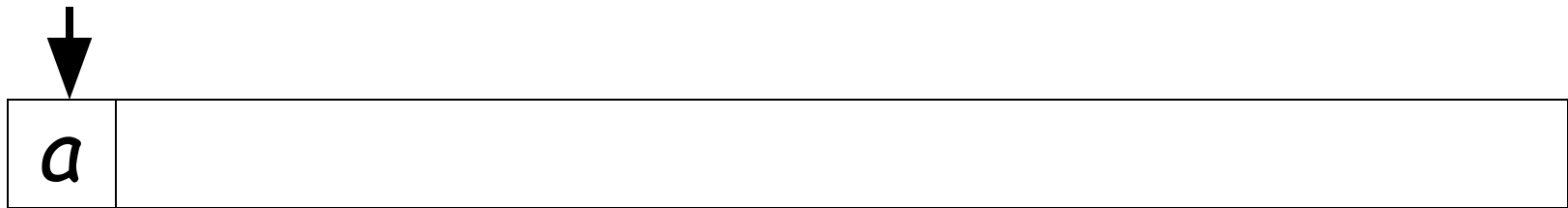
"reject"



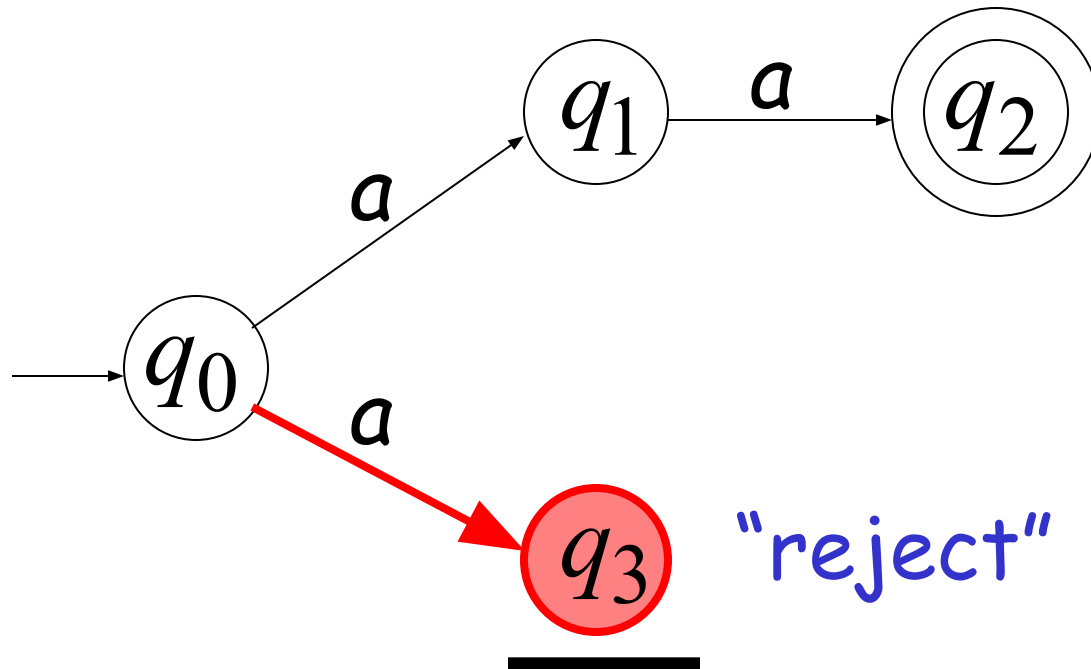
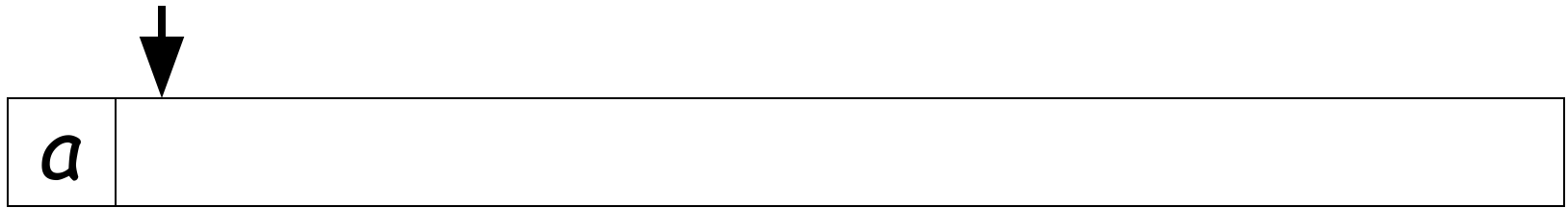
Second Choice



Second Choice



Second Choice



An NFA rejects a string:

when there is no computation of the NFA that accepts the string:

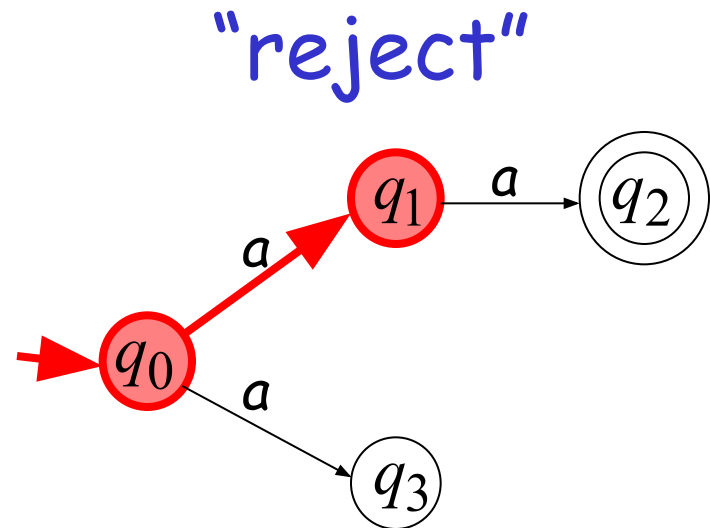
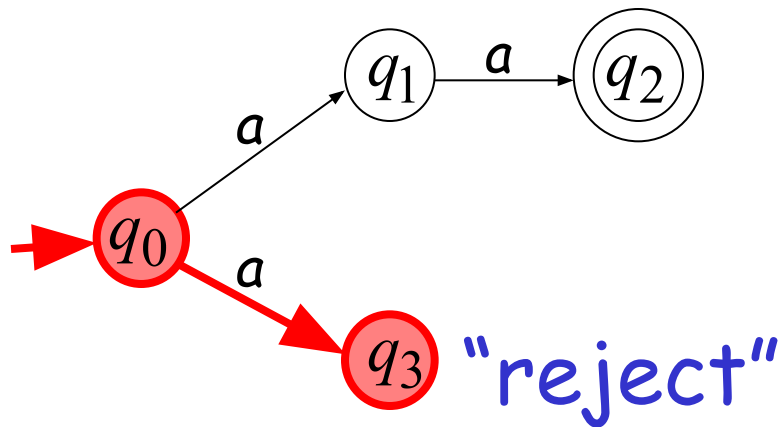
- All the input is consumed and the automaton is in a non final state

OR

- The input cannot be consumed

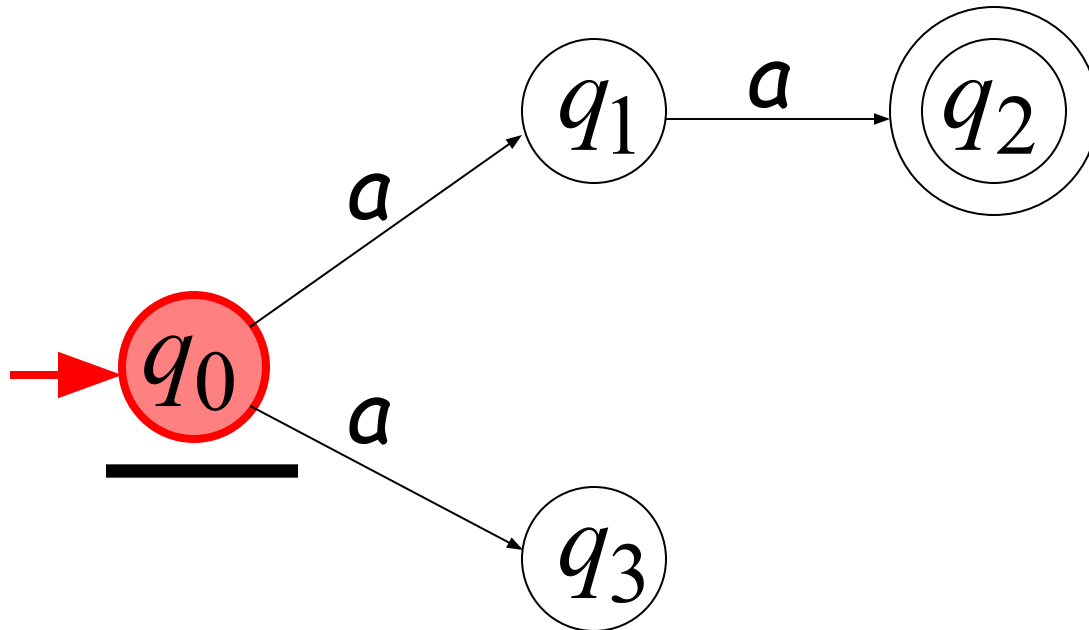
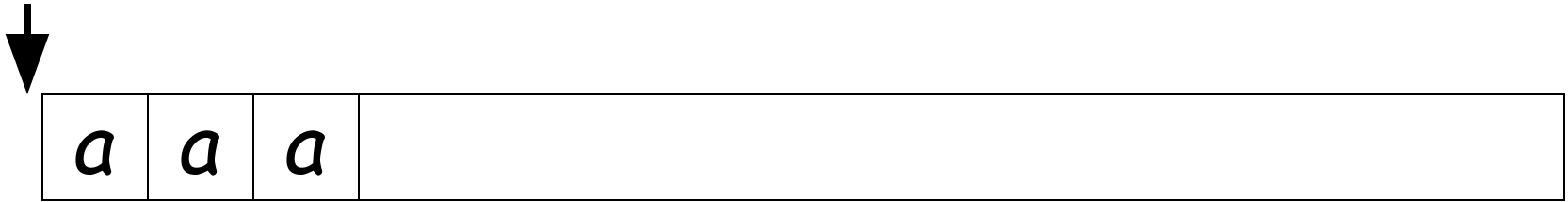
Example

a is rejected by the NFA:

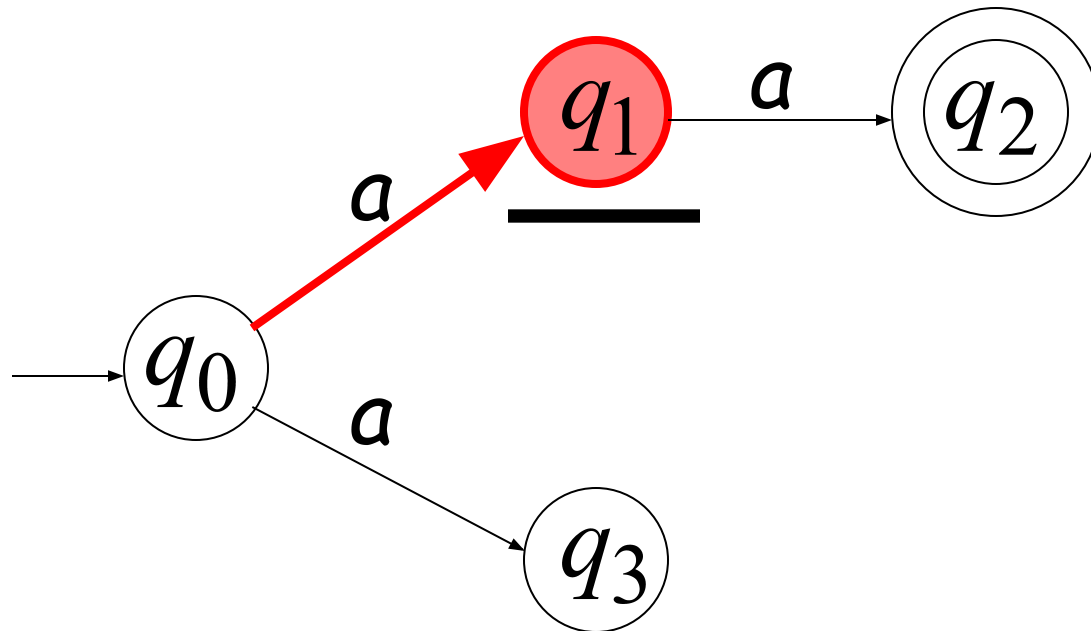
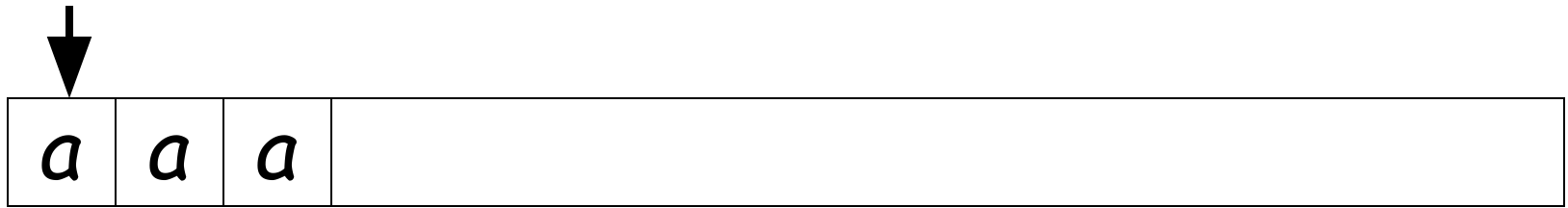


All possible computations lead to rejection

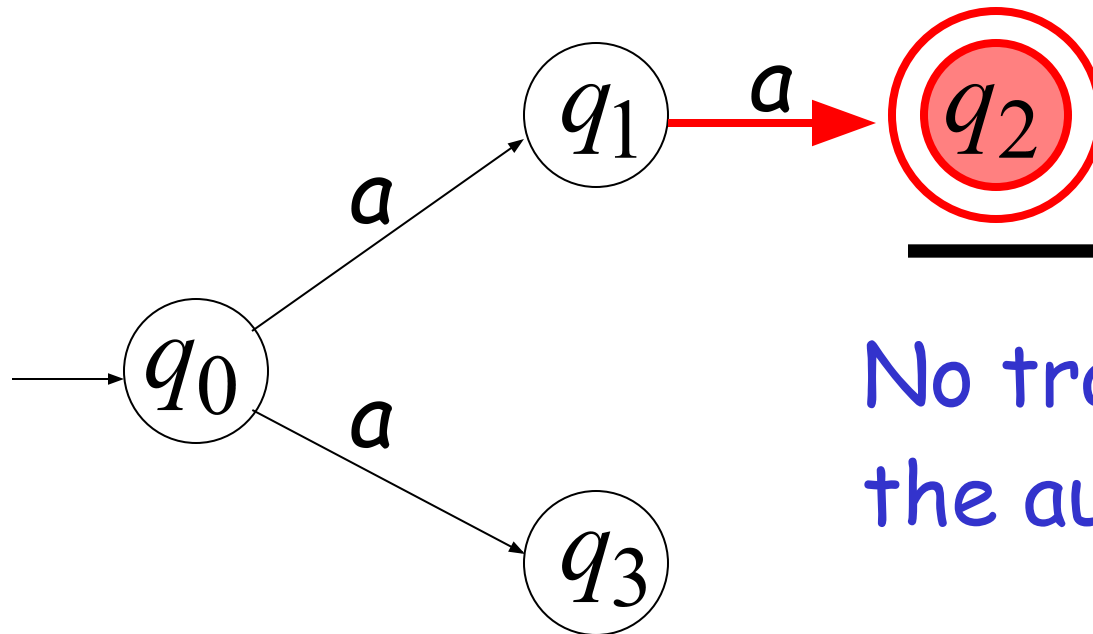
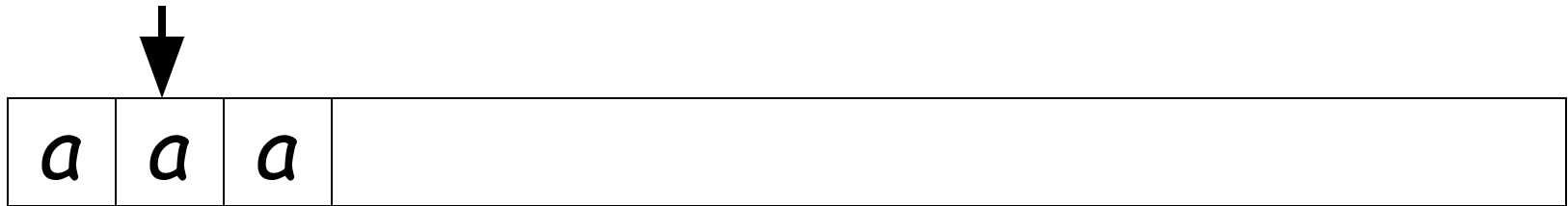
Rejection example



First Choice

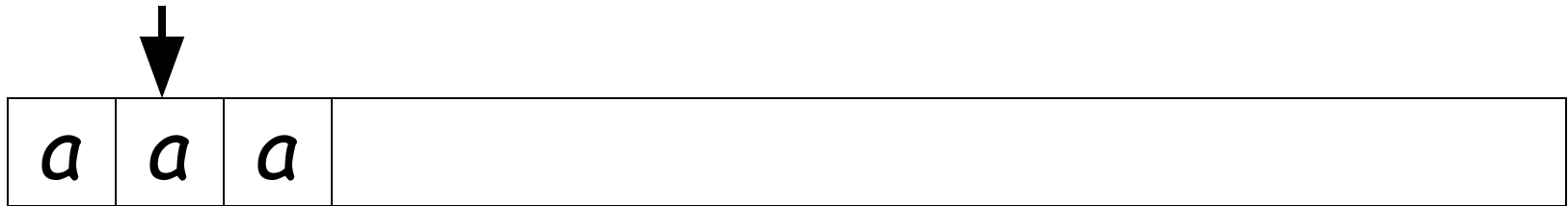


First Choice

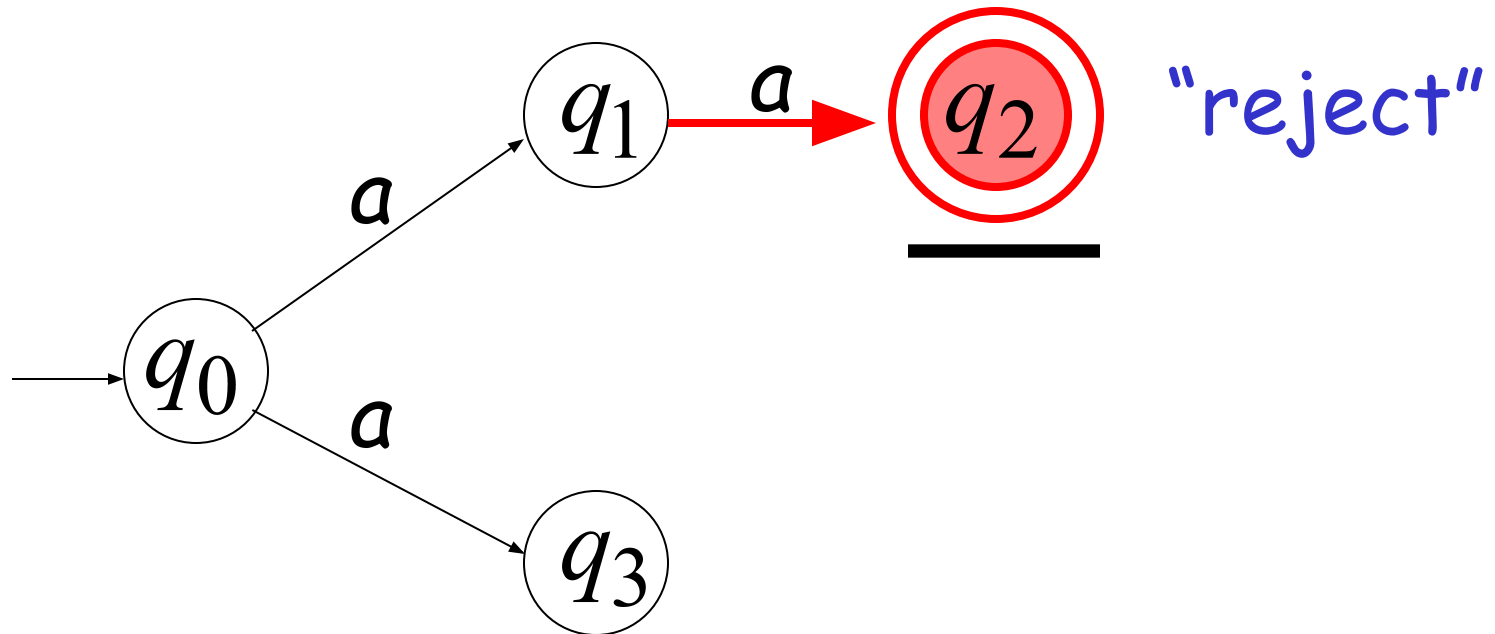


No transition:
the automaton hangs

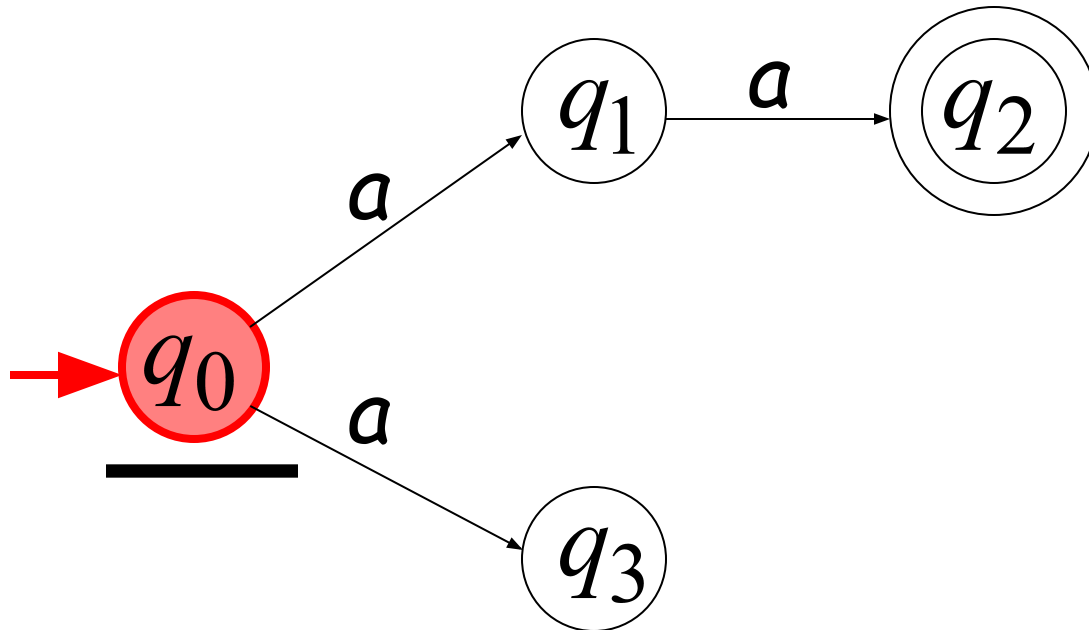
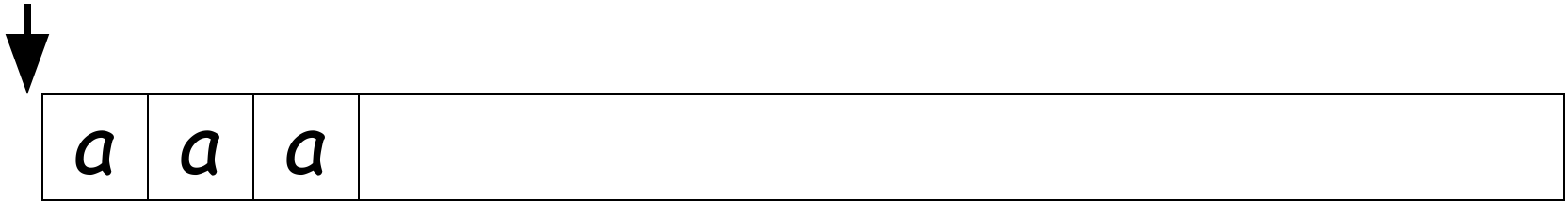
First Choice



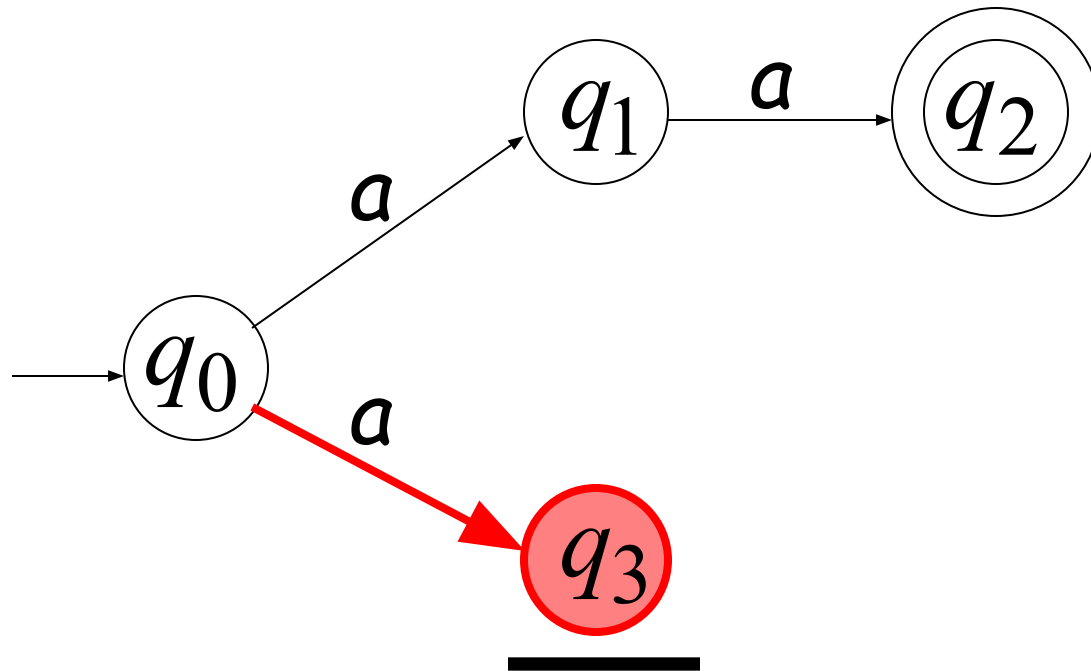
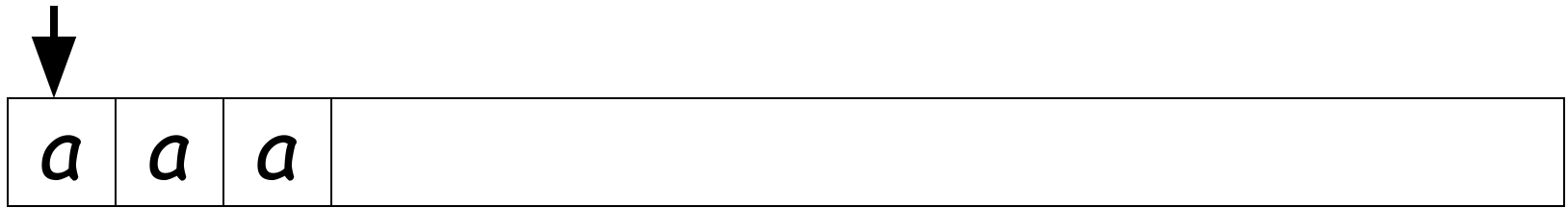
Input cannot be consumed



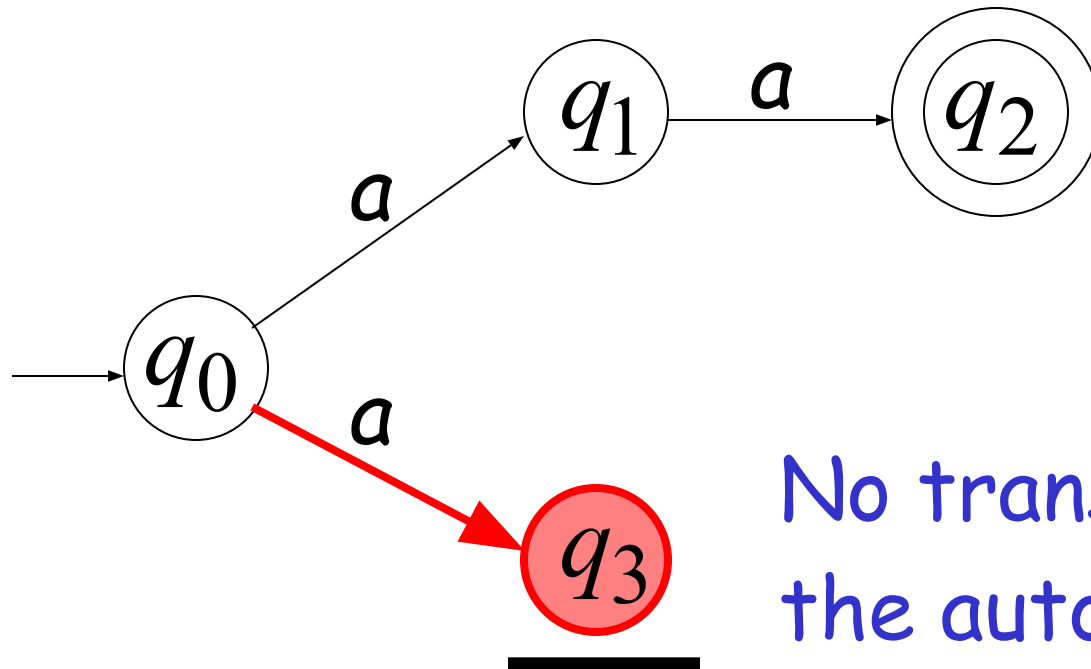
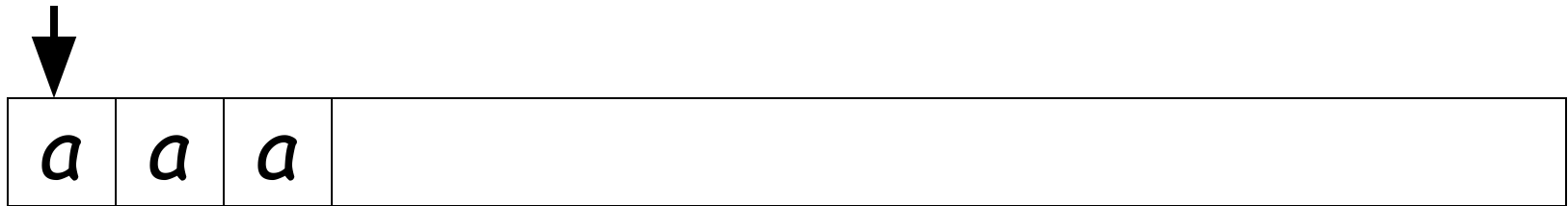
Second Choice



Second Choice

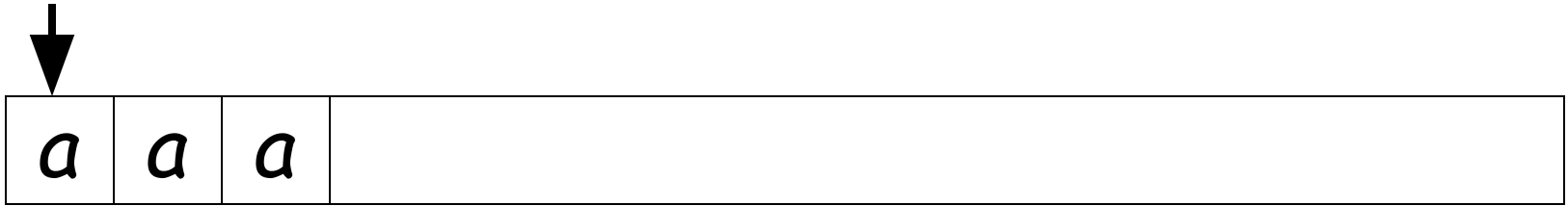


Second Choice

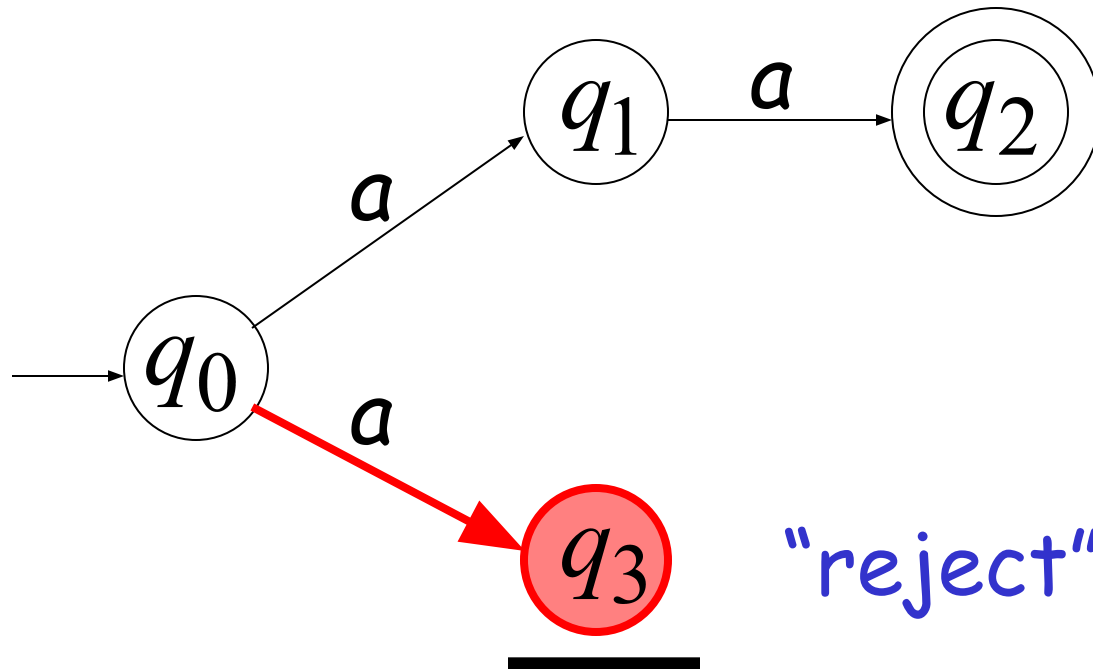


No transition:
the automaton hangs

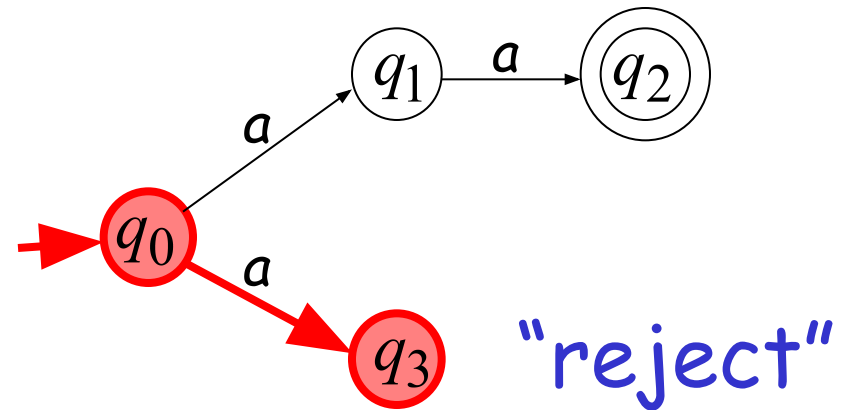
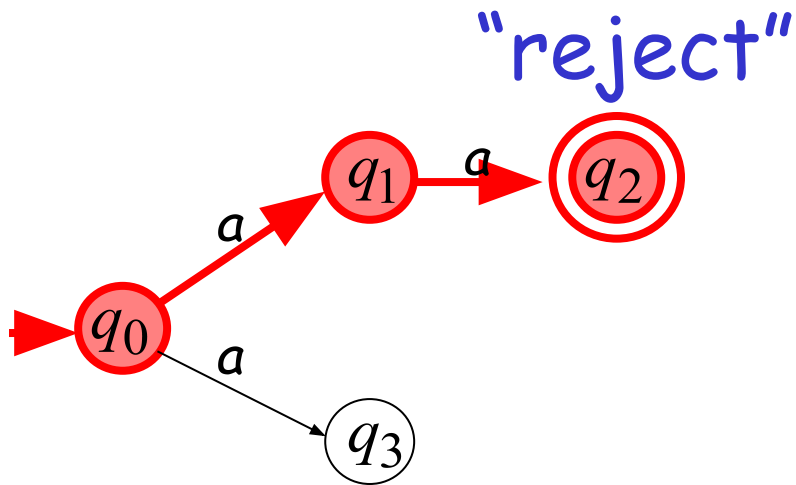
Second Choice



Input cannot be consumed

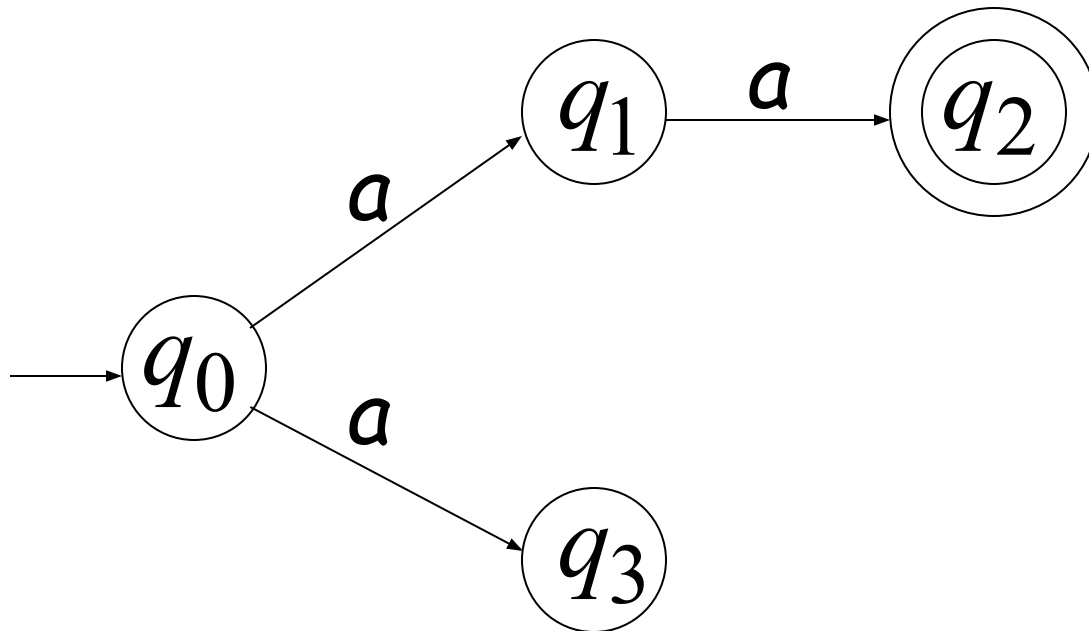


aaa is rejected by the NFA:

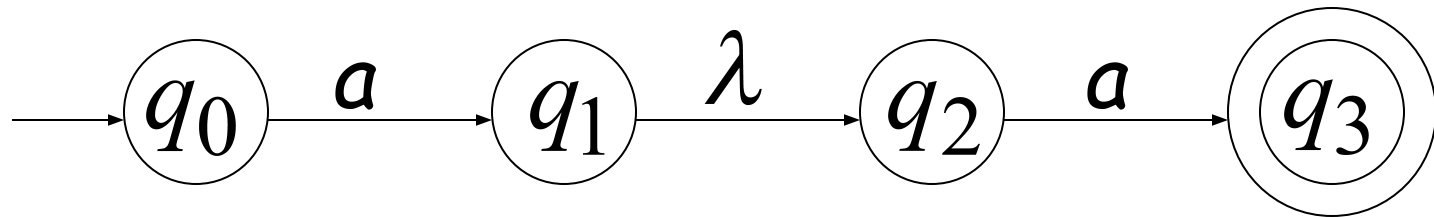


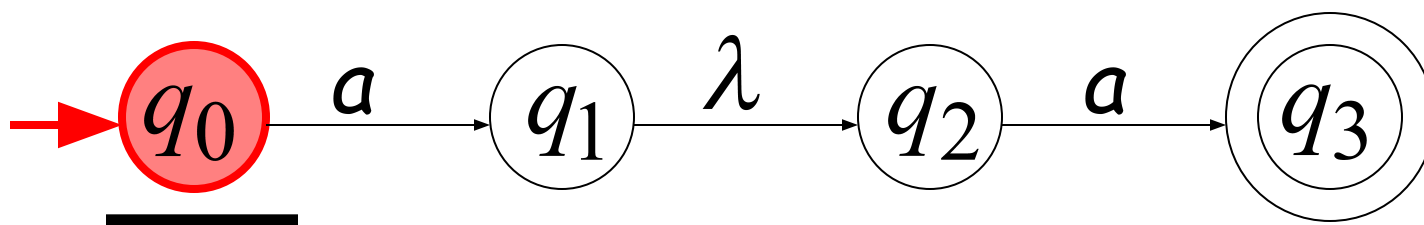
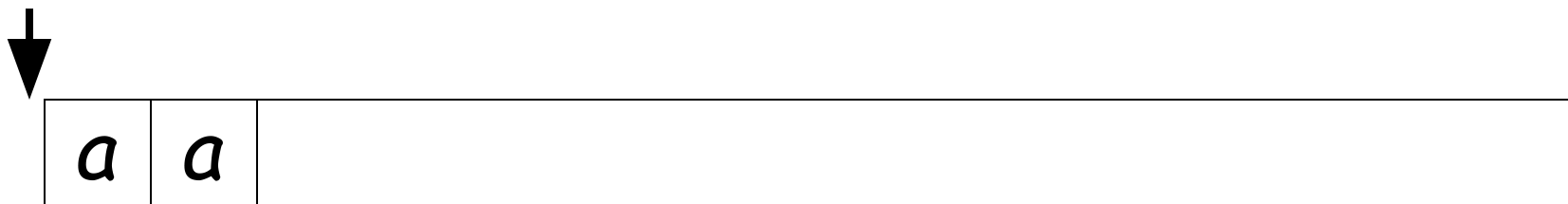
All possible computations lead to rejection

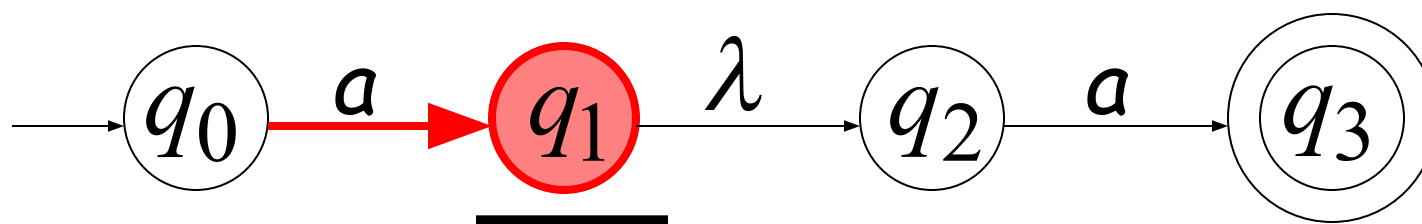
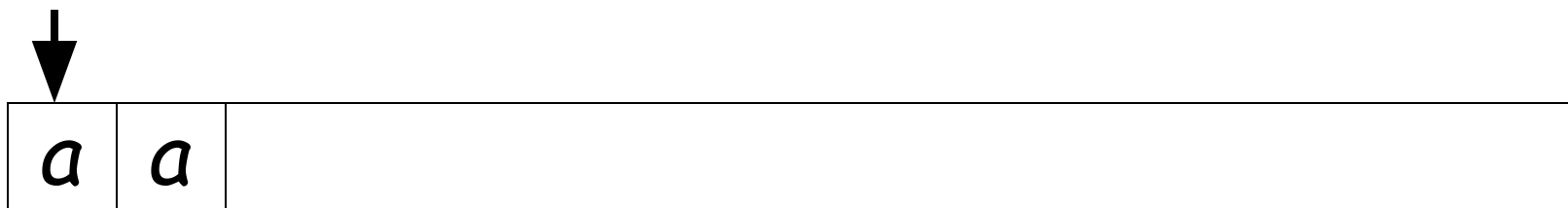
Language accepted: $L = \{aa\}$



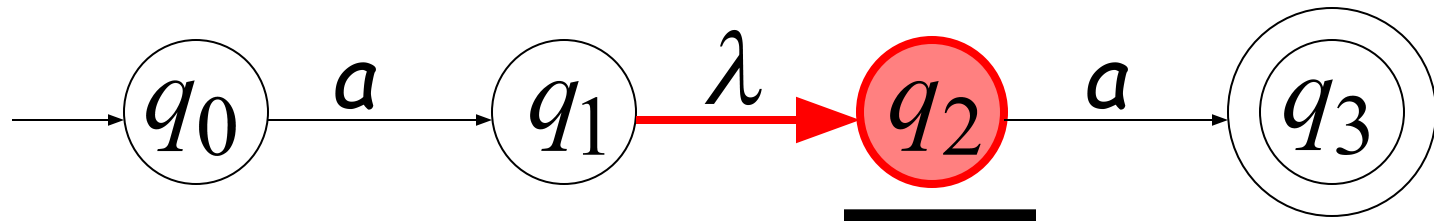
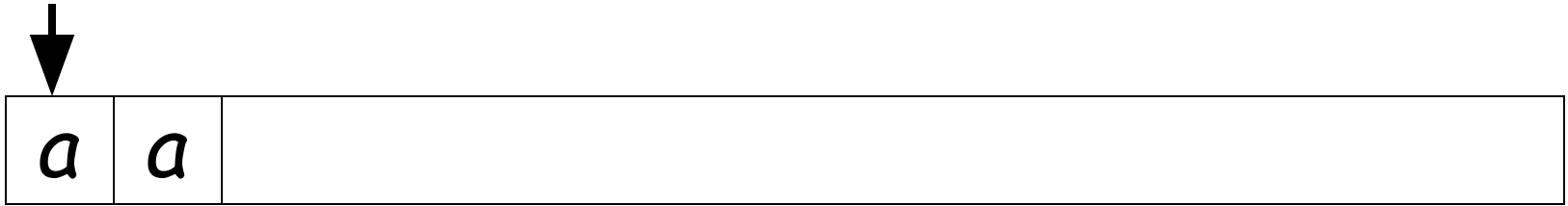
Lambda Transitions

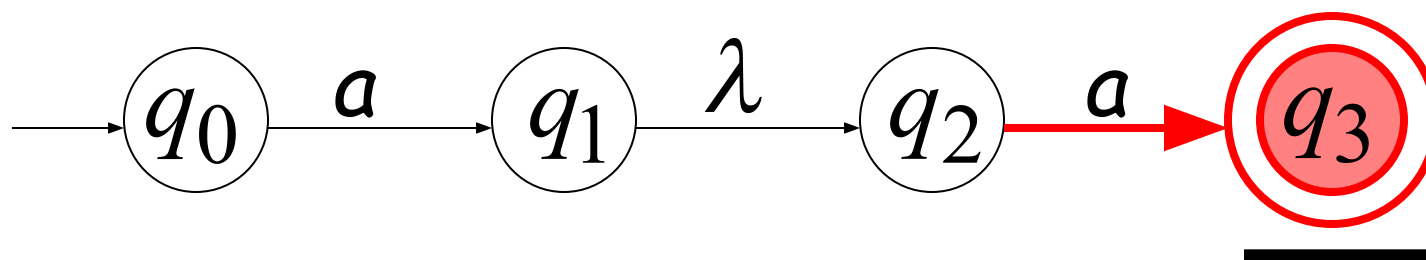
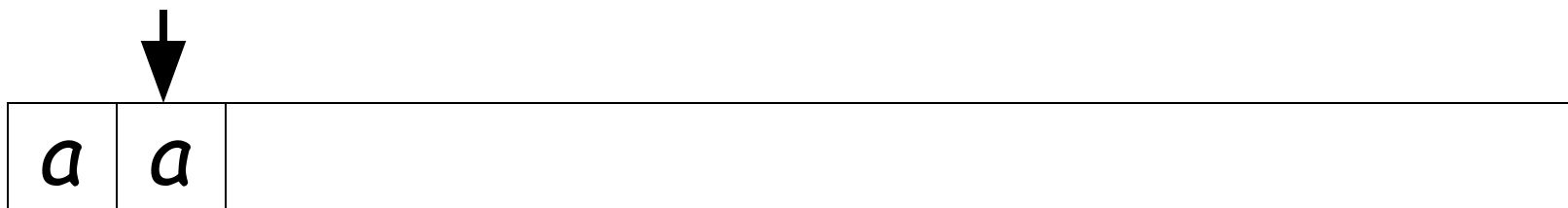




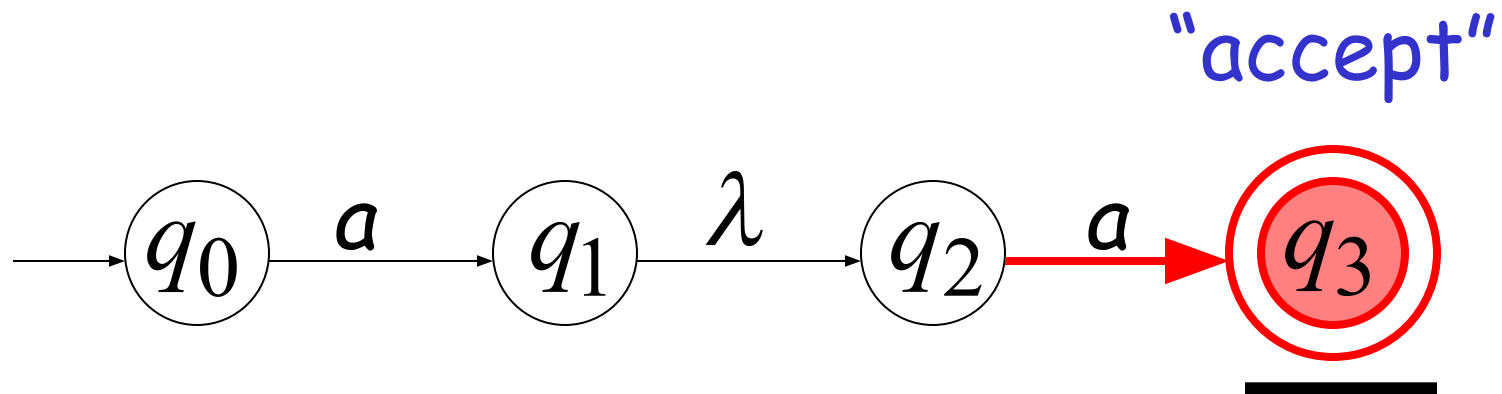
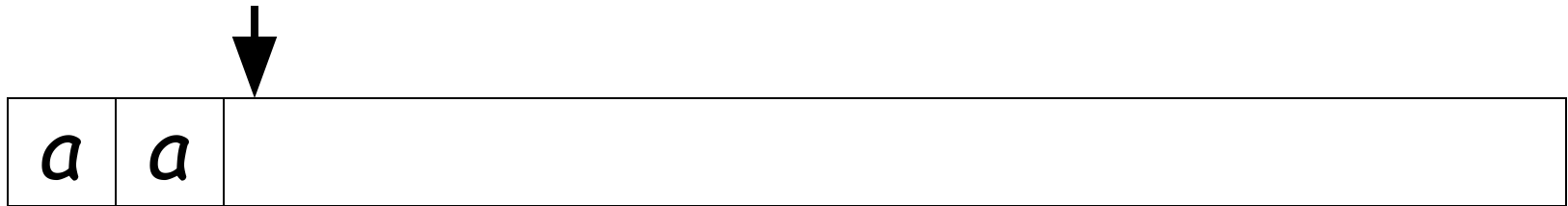


(read head does not move)



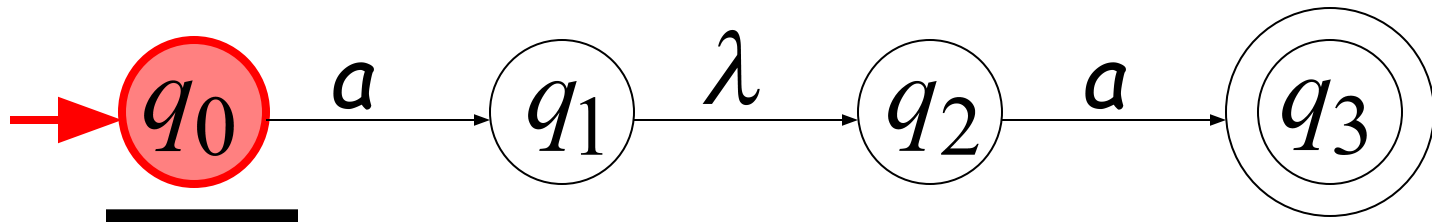
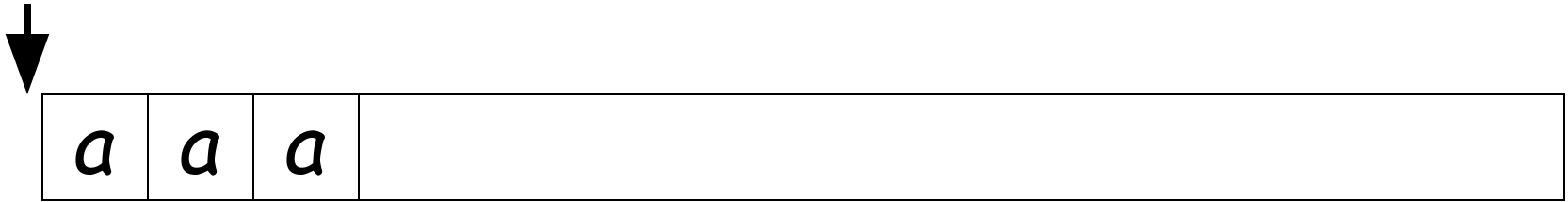


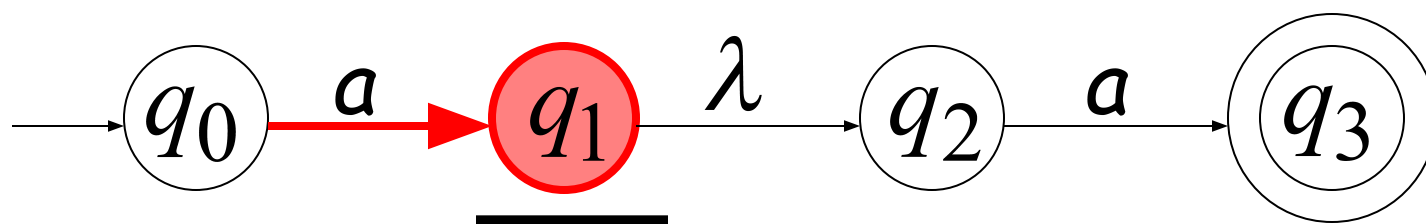
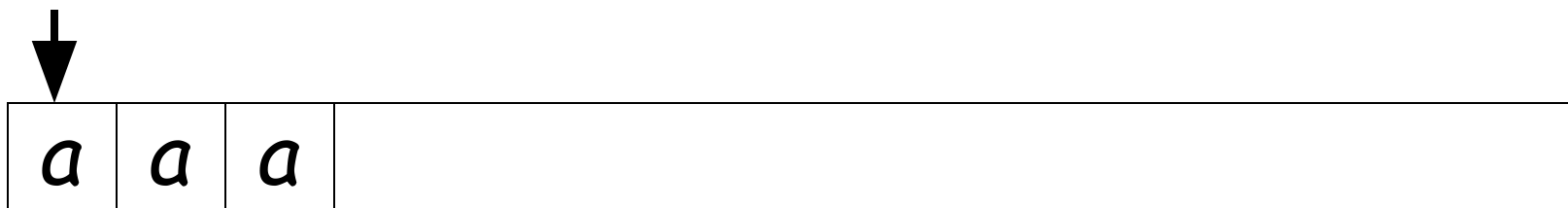
all input is consumed



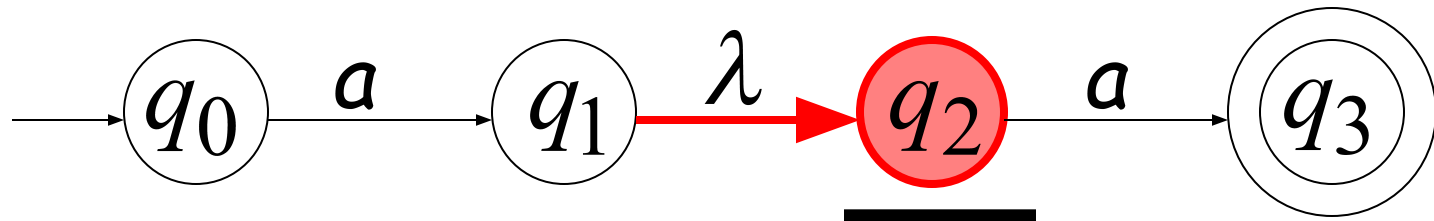
String aa is accepted

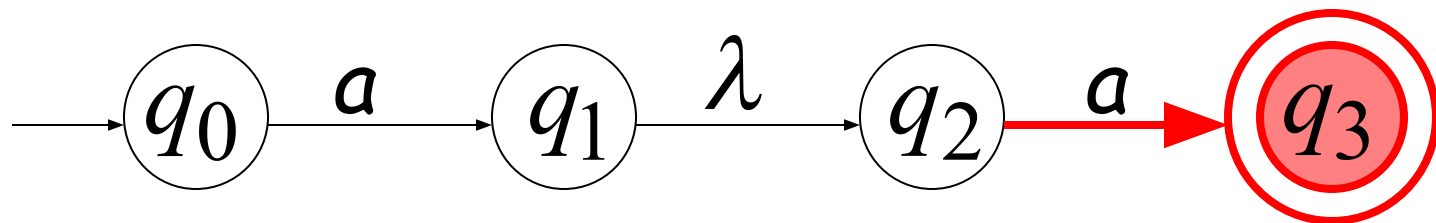
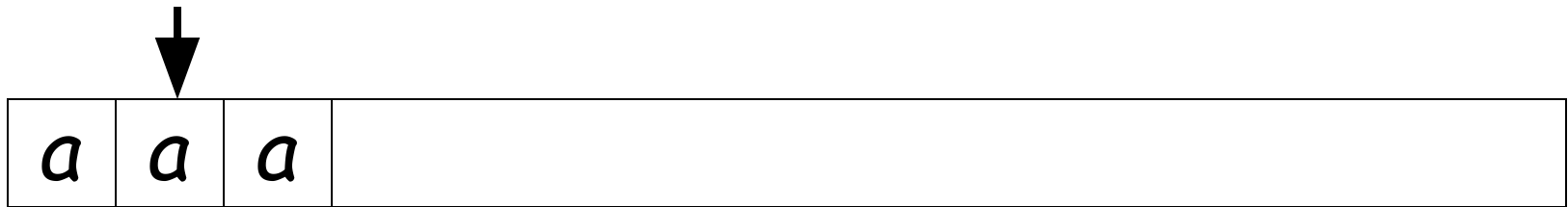
Rejection Example





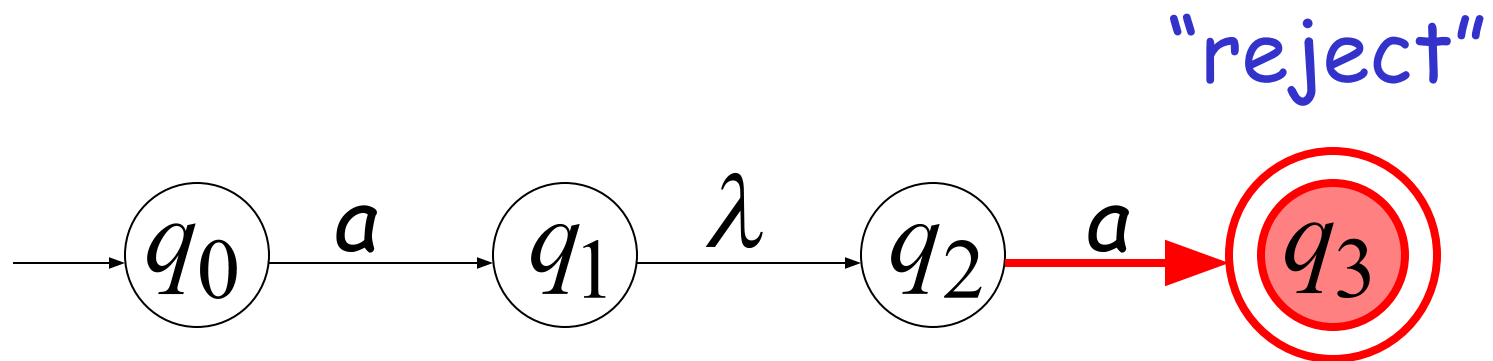
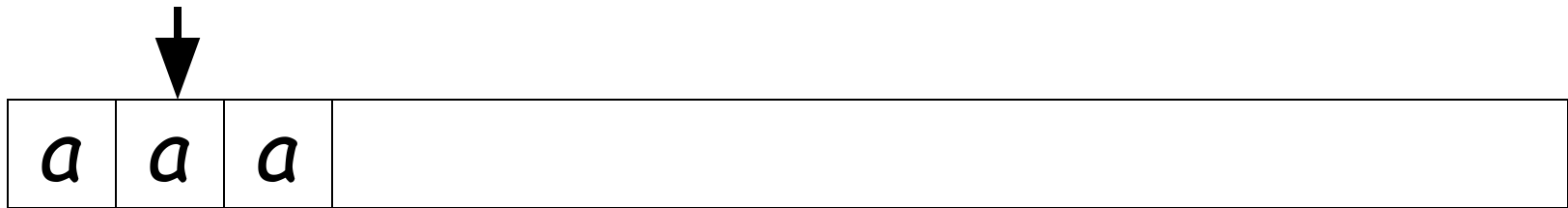
(read head doesn't move)





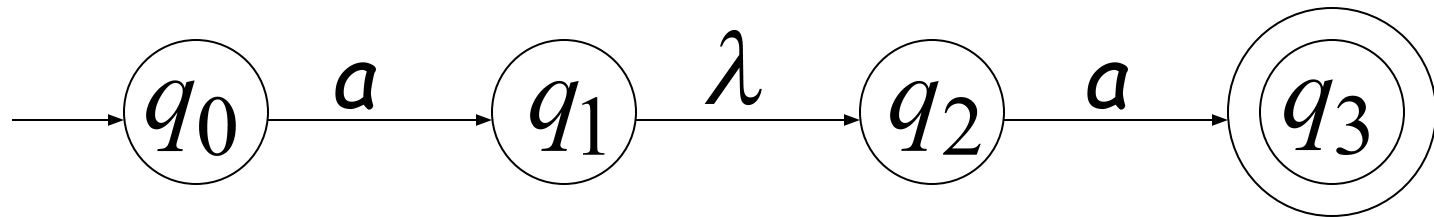
No transition:
the automaton hangs

Input cannot be consumed

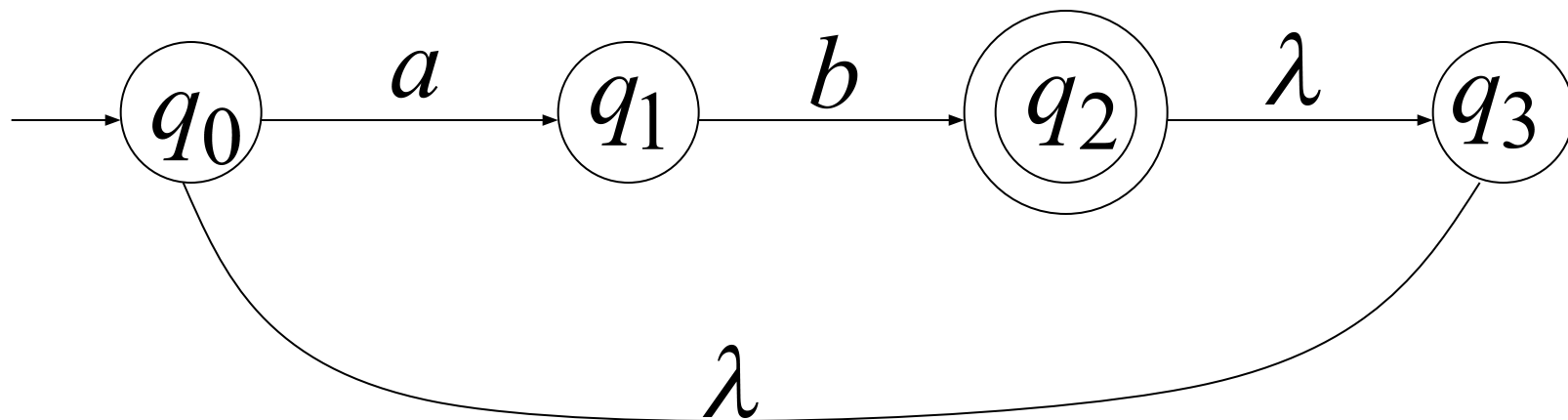


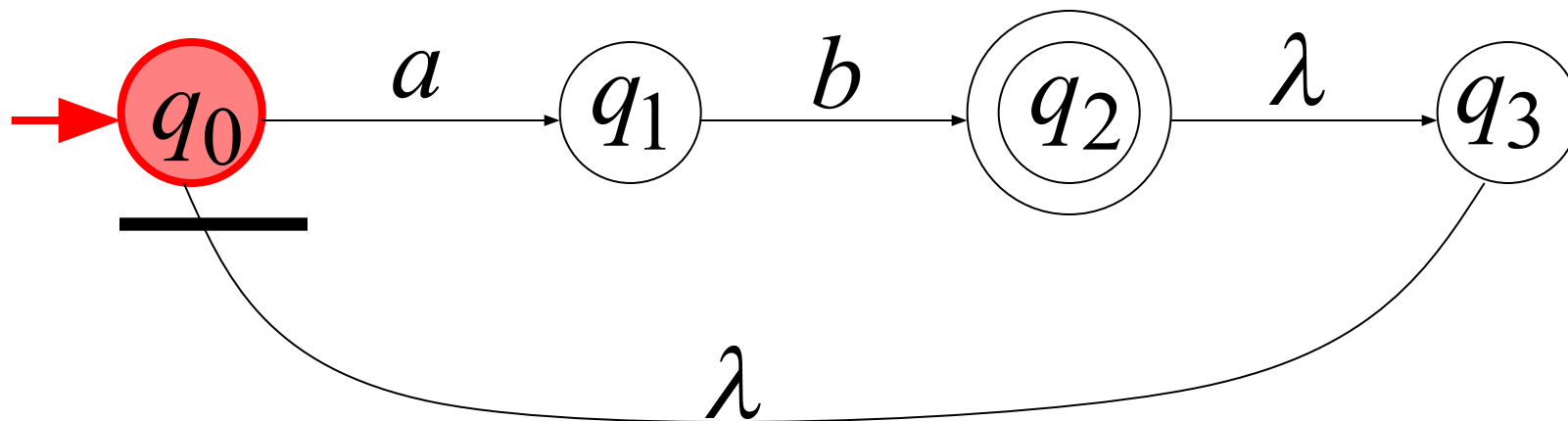
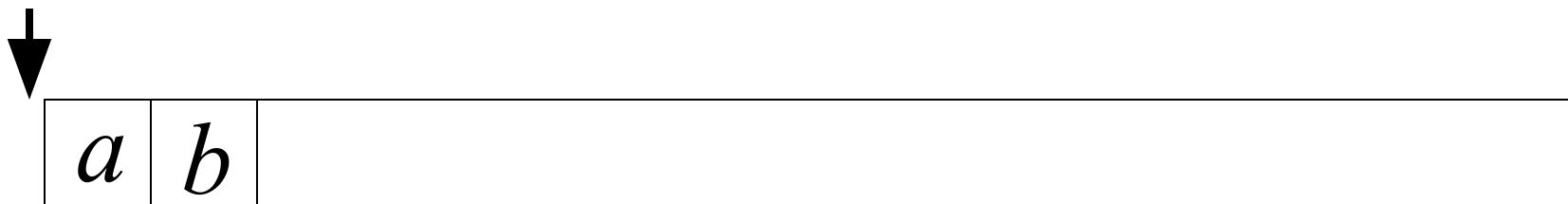
String **aaa** is rejected

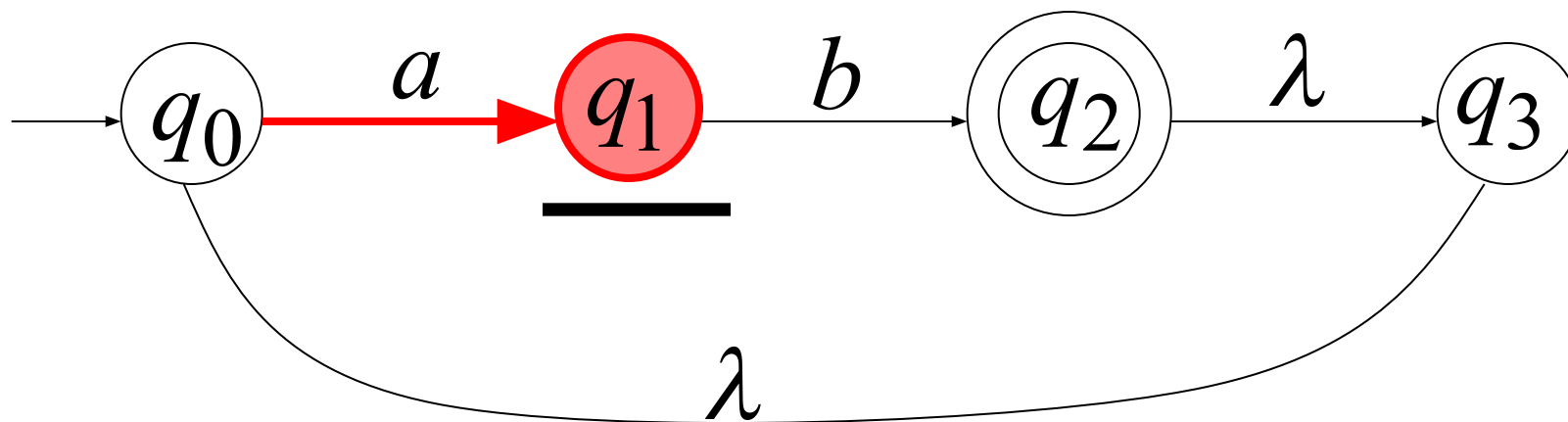
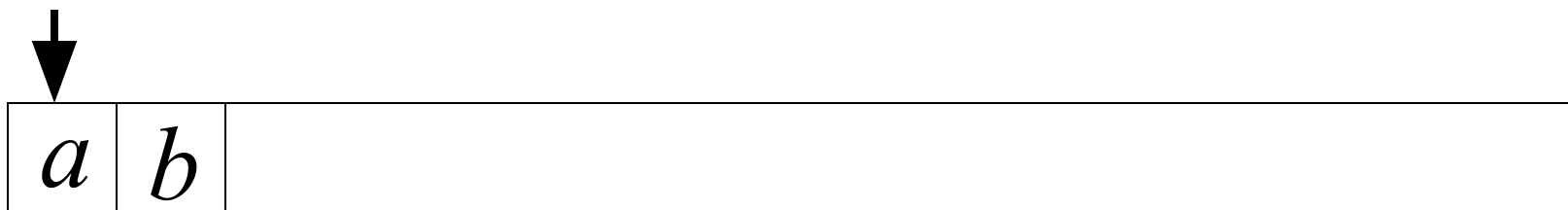
Language accepted: $L = \{aa\}$

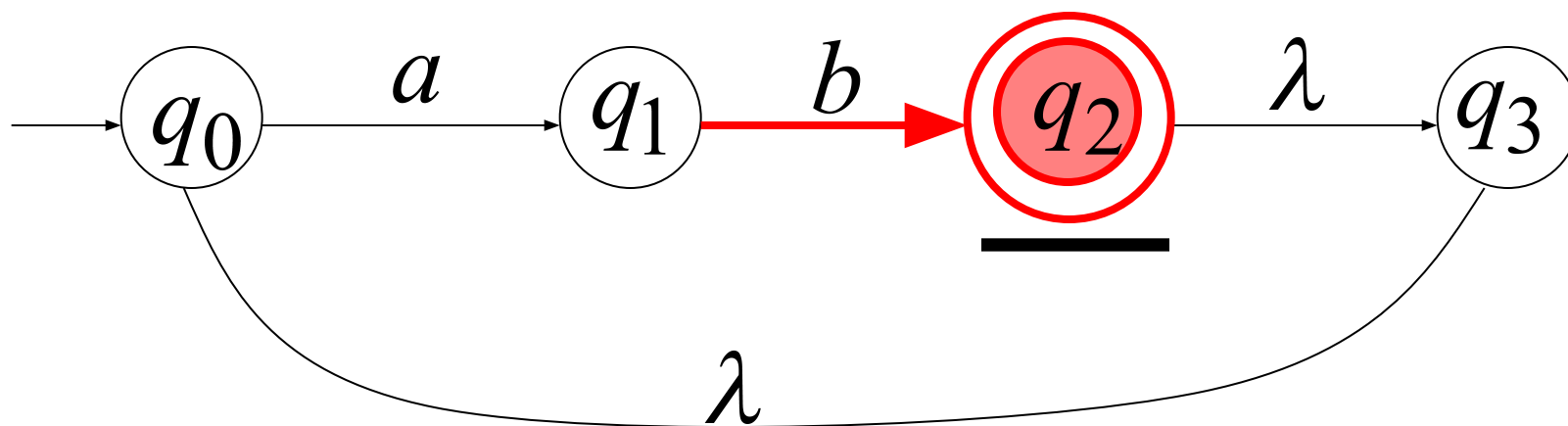
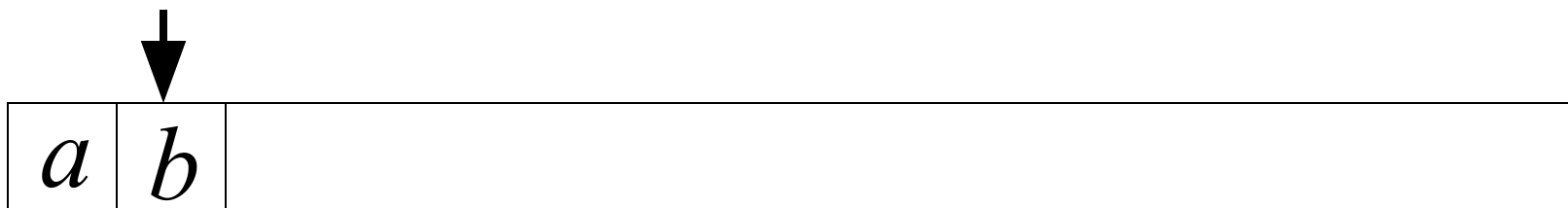


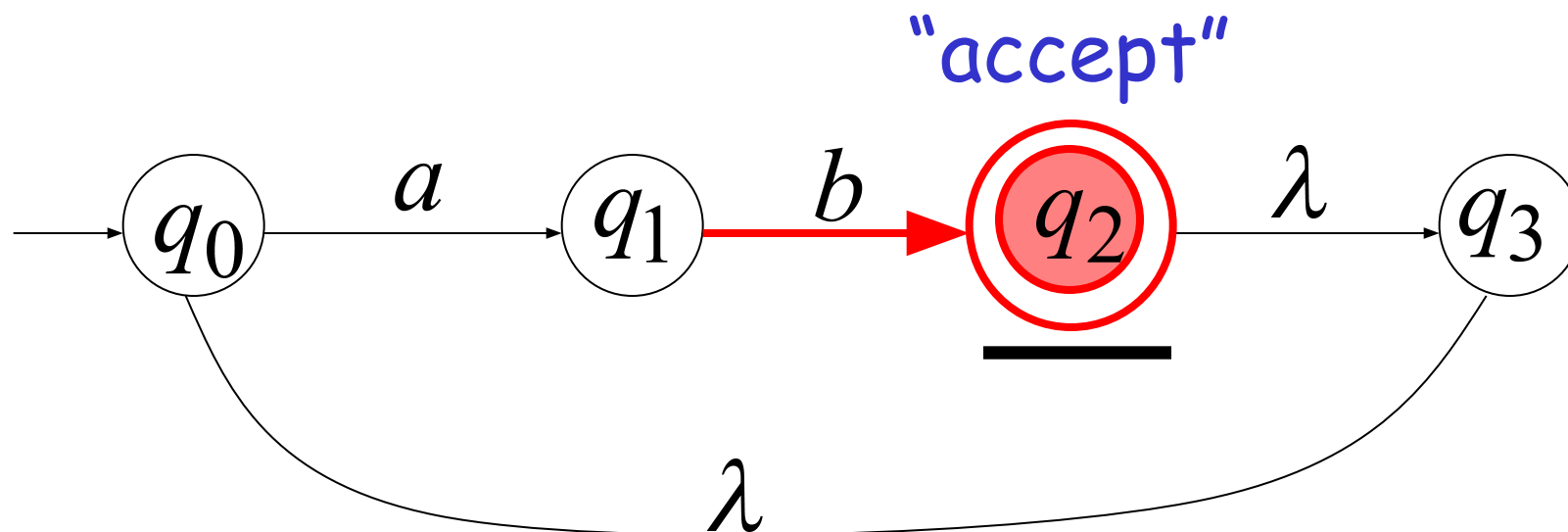
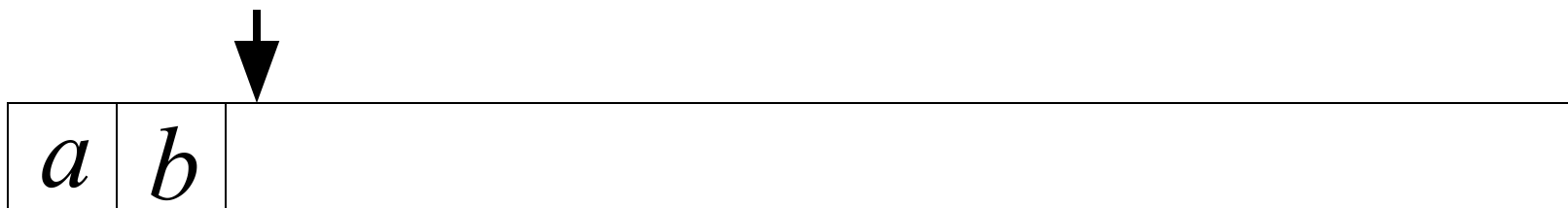
Another NFA Example



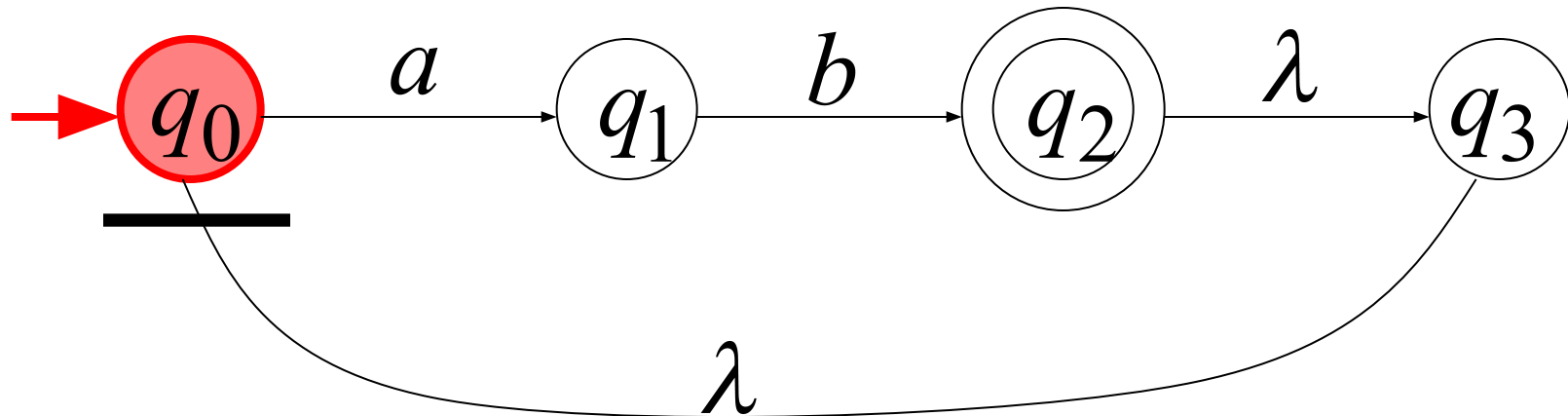
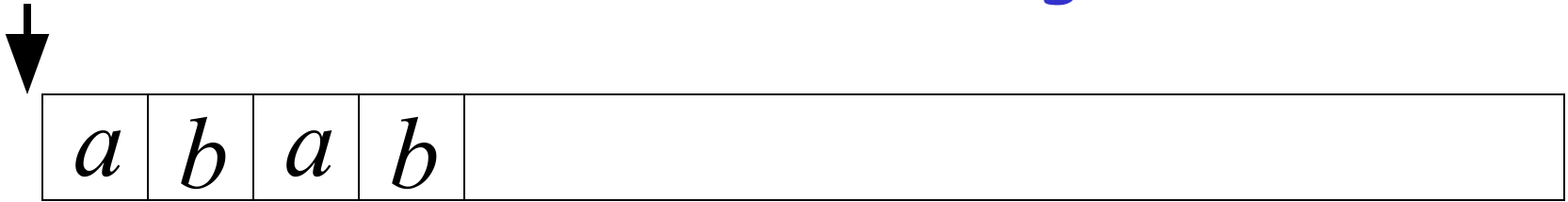


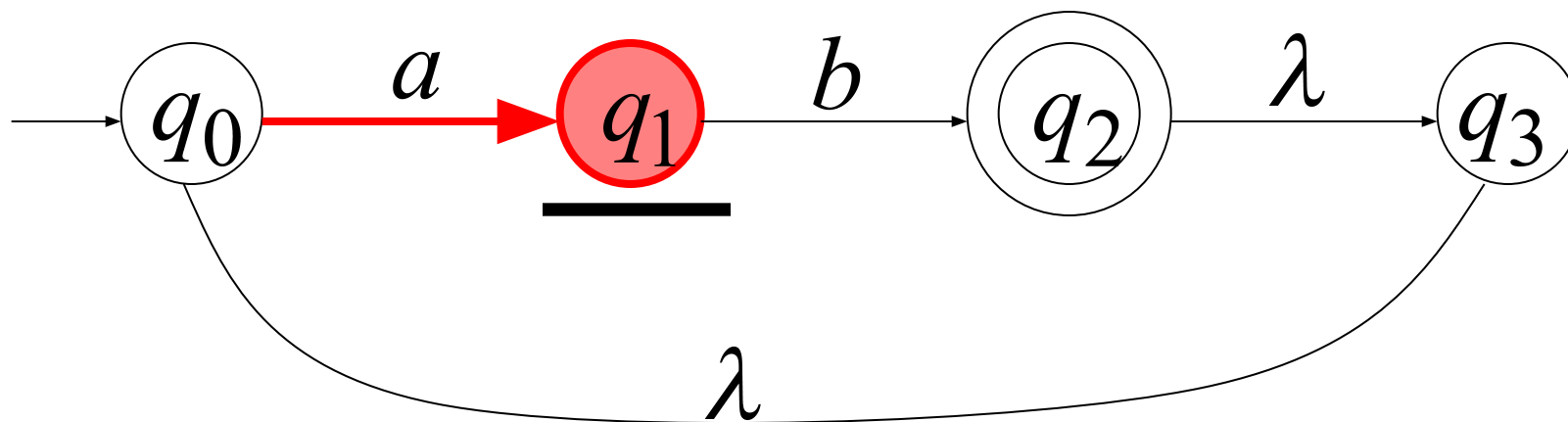
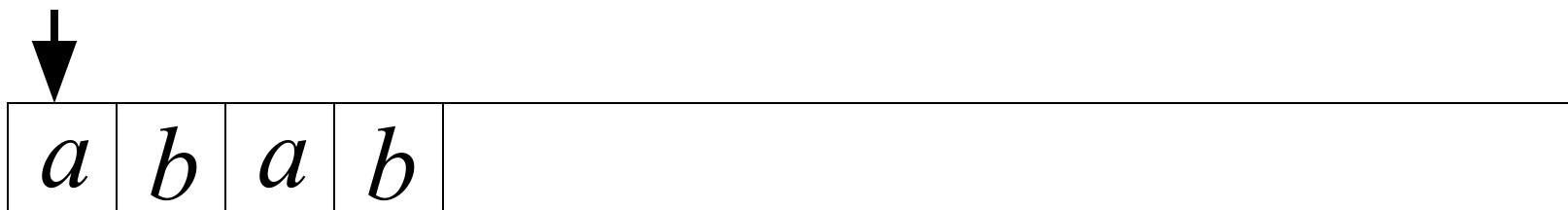


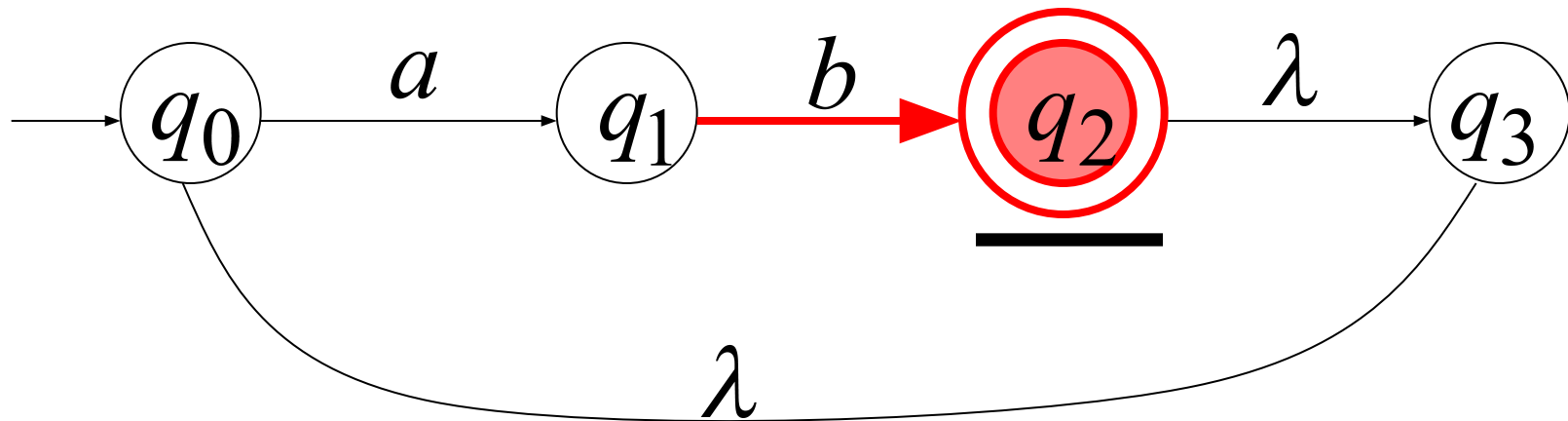
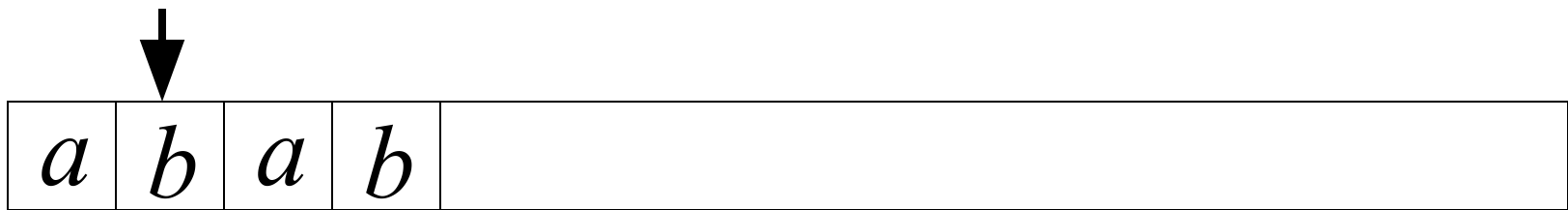


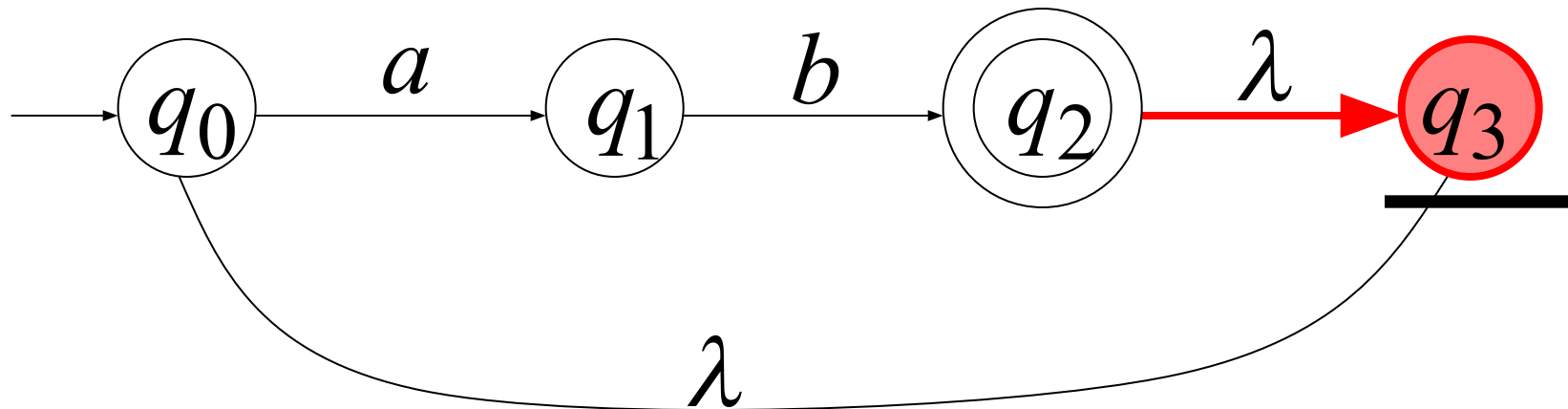
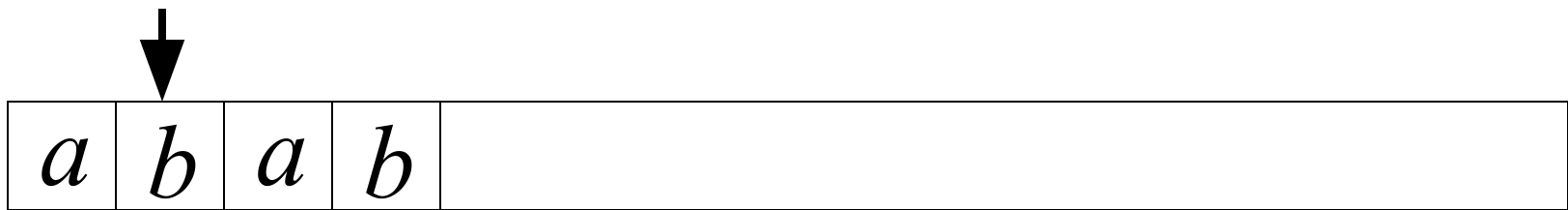


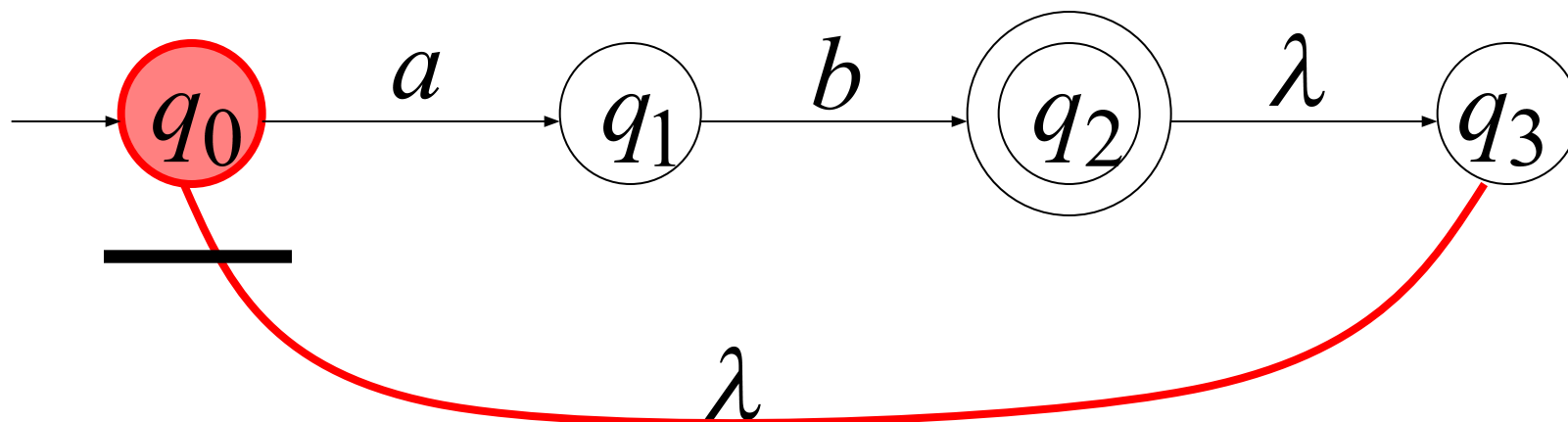
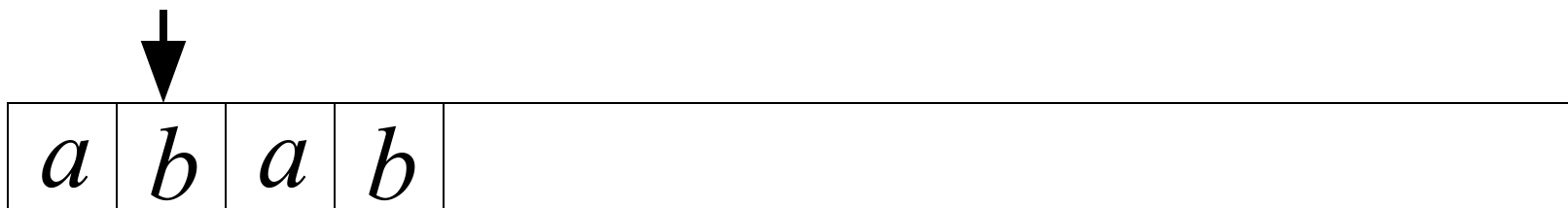
Another String

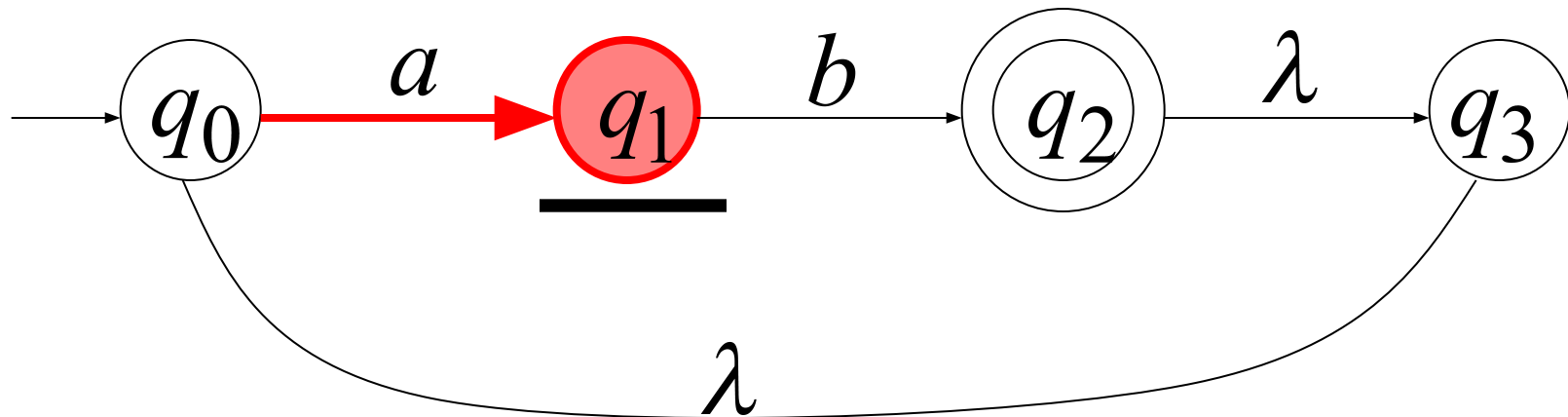
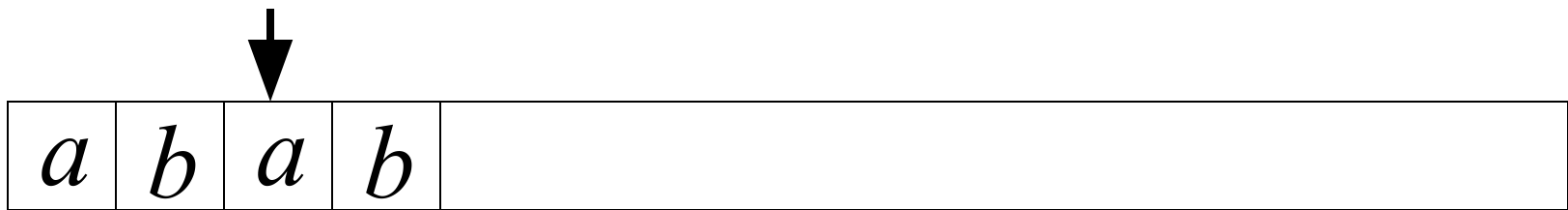


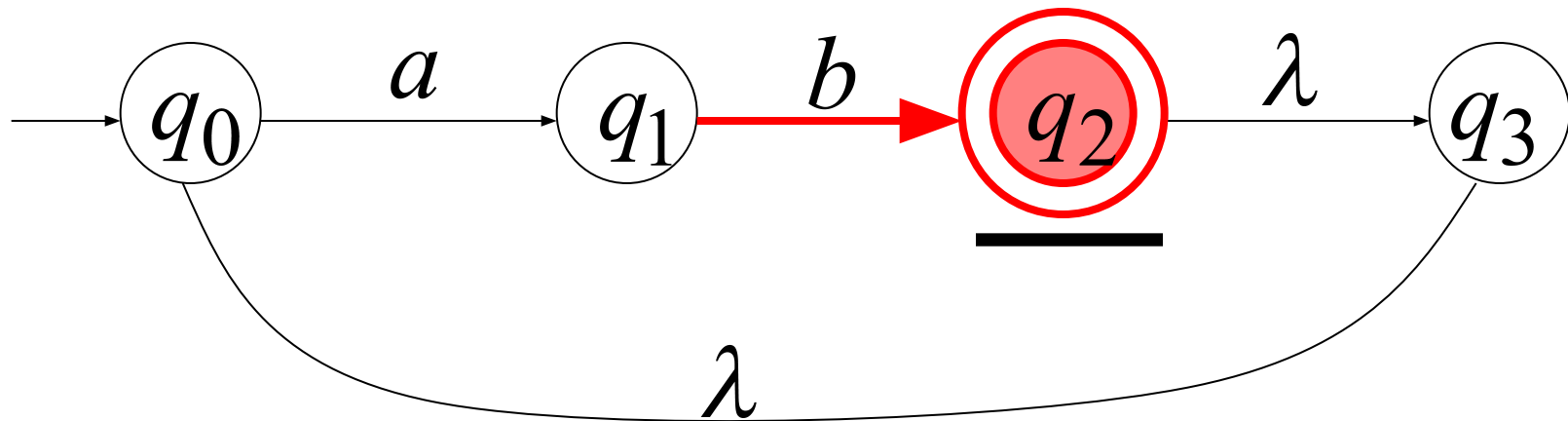
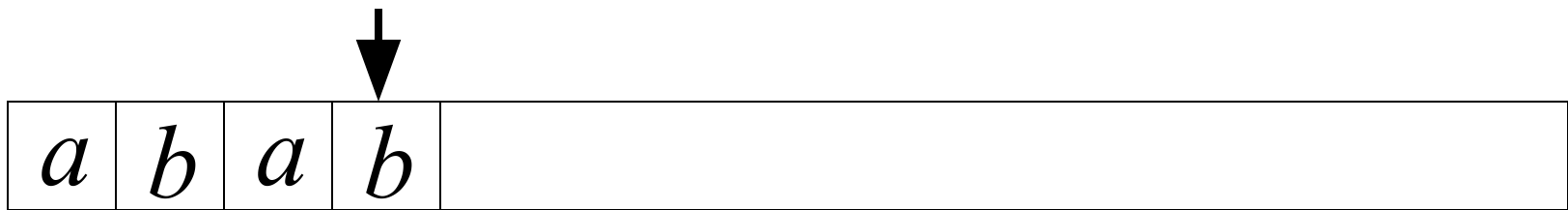


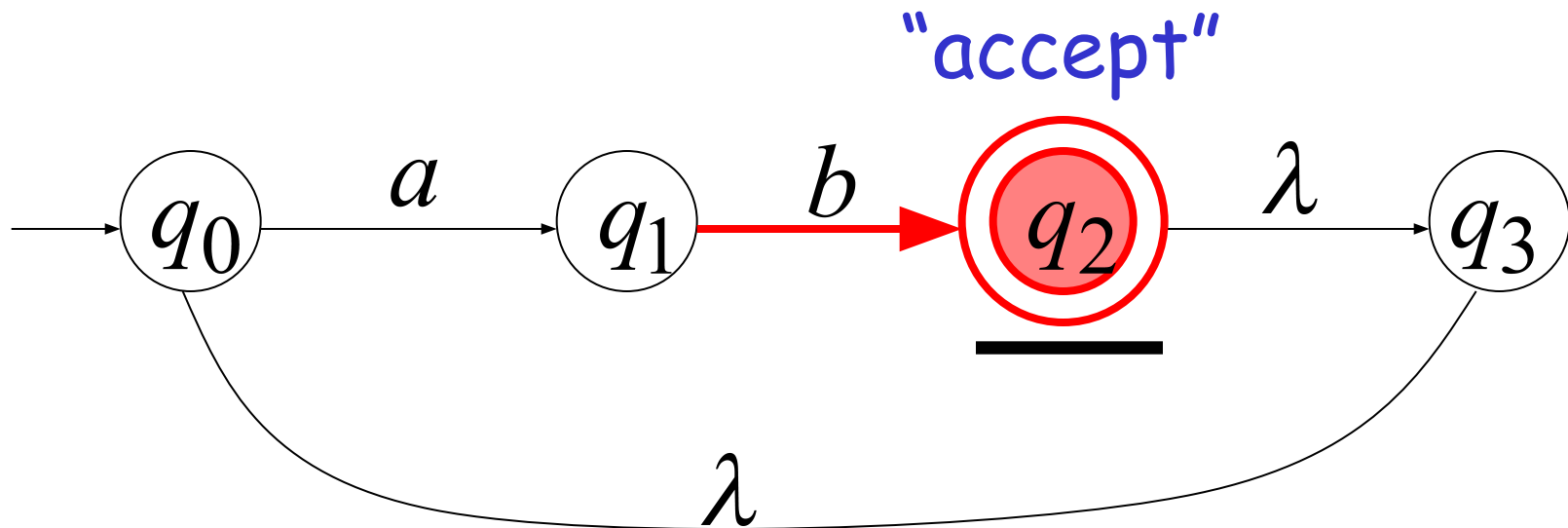
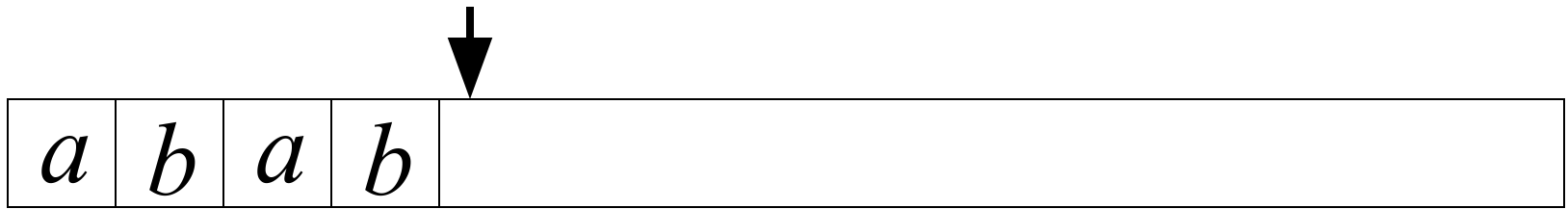






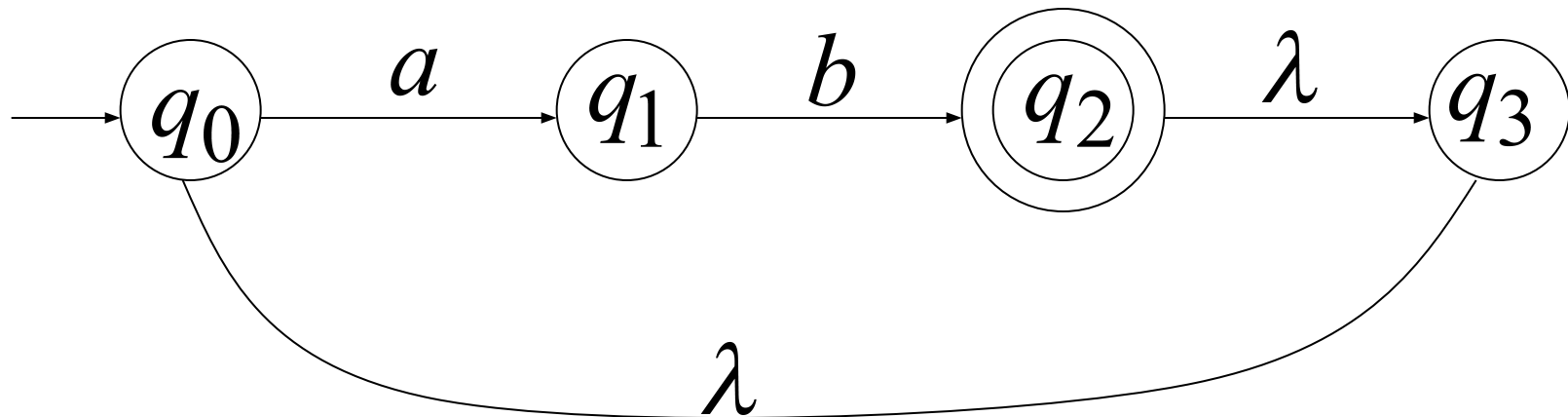




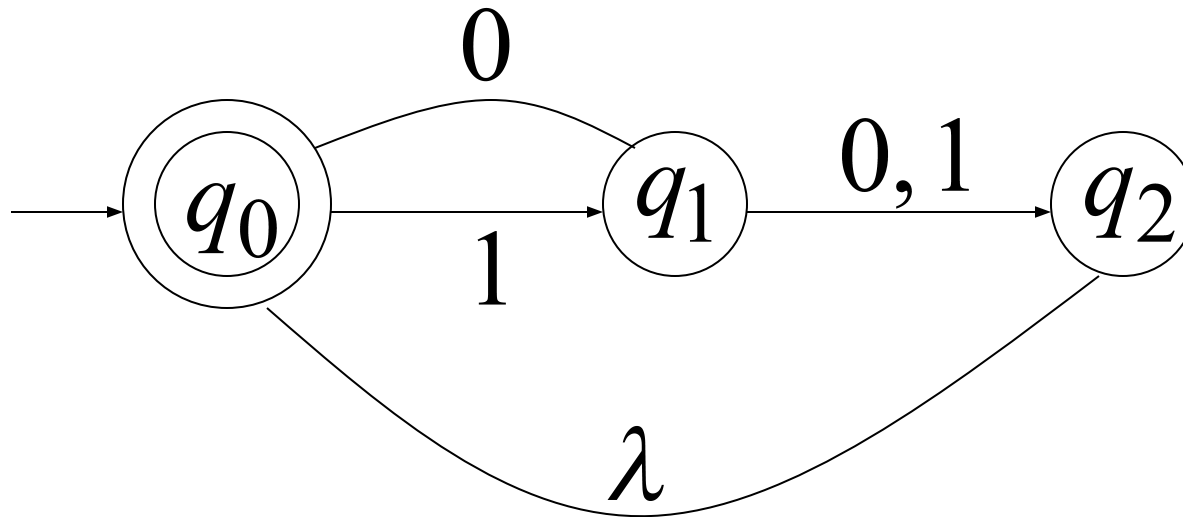


Language accepted

$$L = \{ab, abab, ababab, \dots\}$$
$$= \{ab\}^+$$

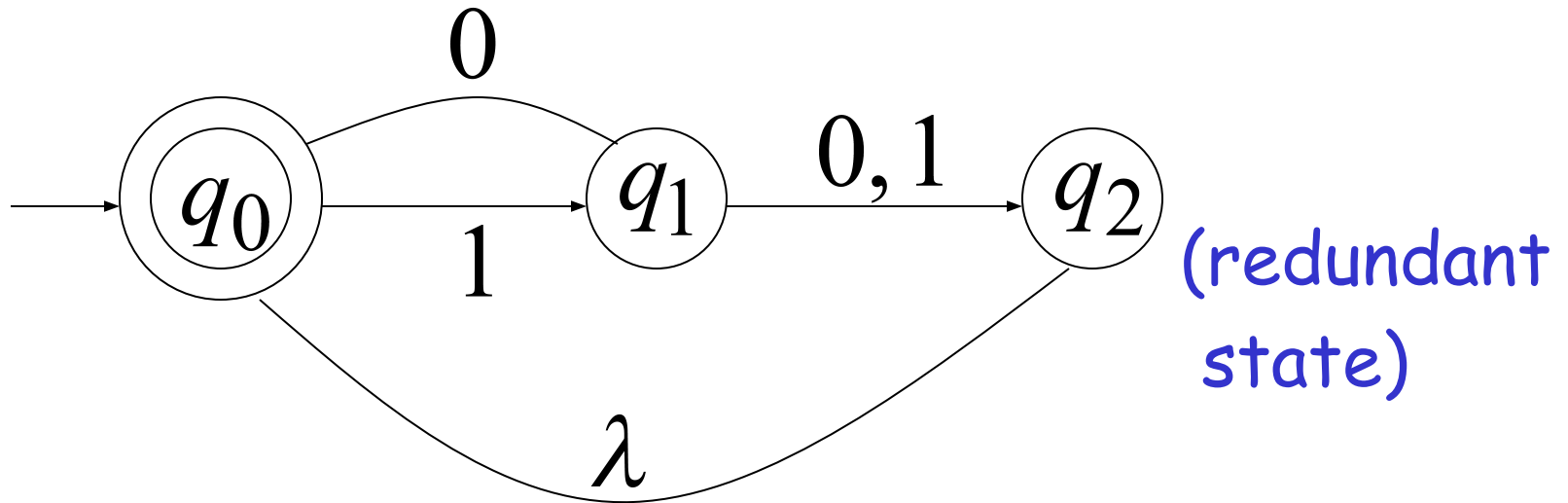


Another NFA Example



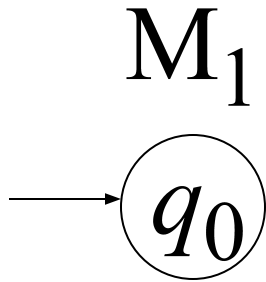
Language accepted

$$L(M) = \{\lambda, 10, 1010, 101010, \dots\}$$
$$= \{10\}^*$$

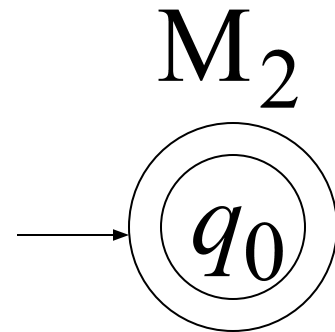


Remarks:

- The λ symbol never appears on the input tape
- Simple automata:



$$L(M_1) = \{\}$$



$$L(M_2) = \{\lambda\}$$

Formal Definition of NFAs

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q : Set of states, i.e. $\{q_0, q_1, q_2\}$

Σ : Input alphabet, i.e. $\{a, b\}$

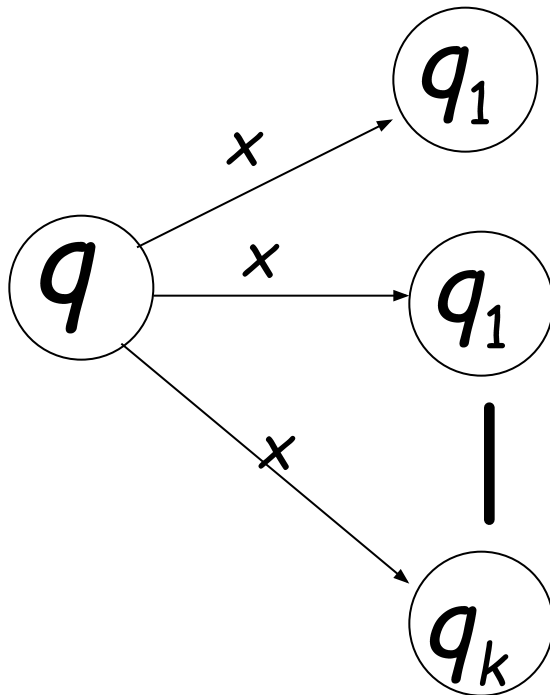
δ : Transition function

q_0 : Initial state

F : Final states

Transition Function δ

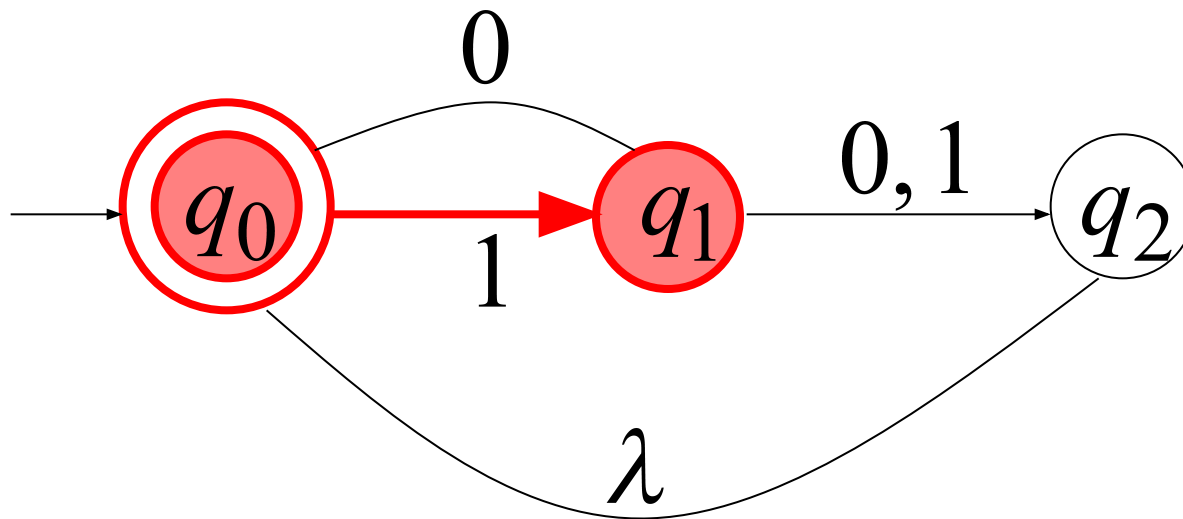
$$\delta(q, x) = \{q_1, q_2, \dots, q_k\}$$



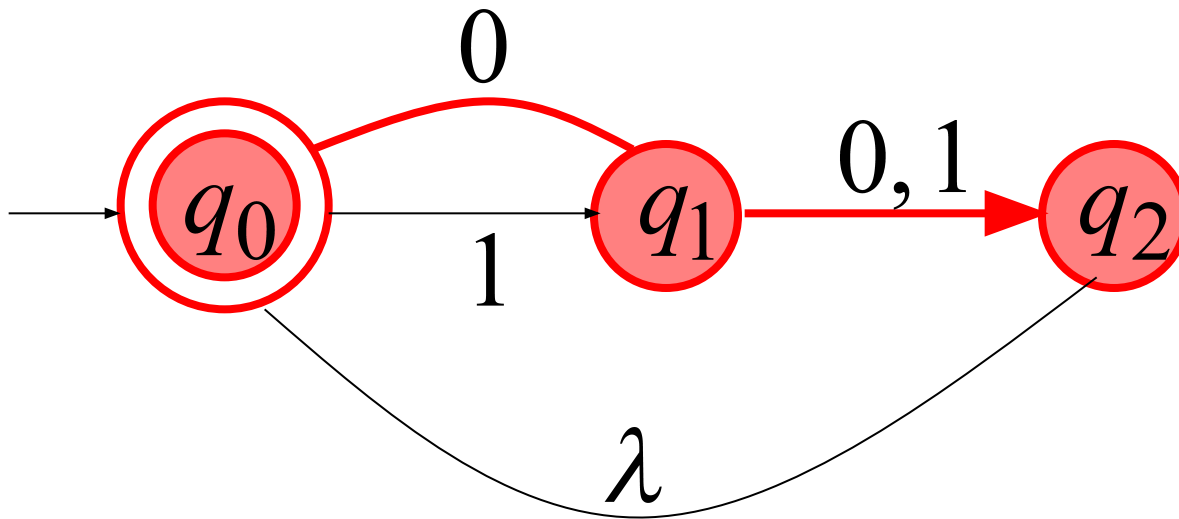
resulting states with
following **one** transition
with symbol x

Transition Function δ

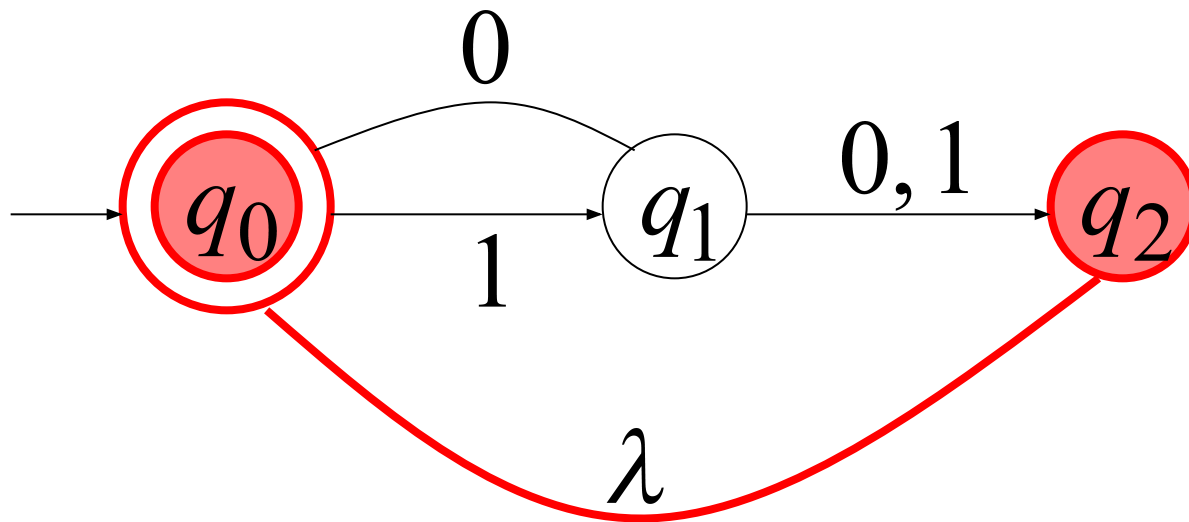
$$\delta(q_0, 1) = \{q_1\}$$



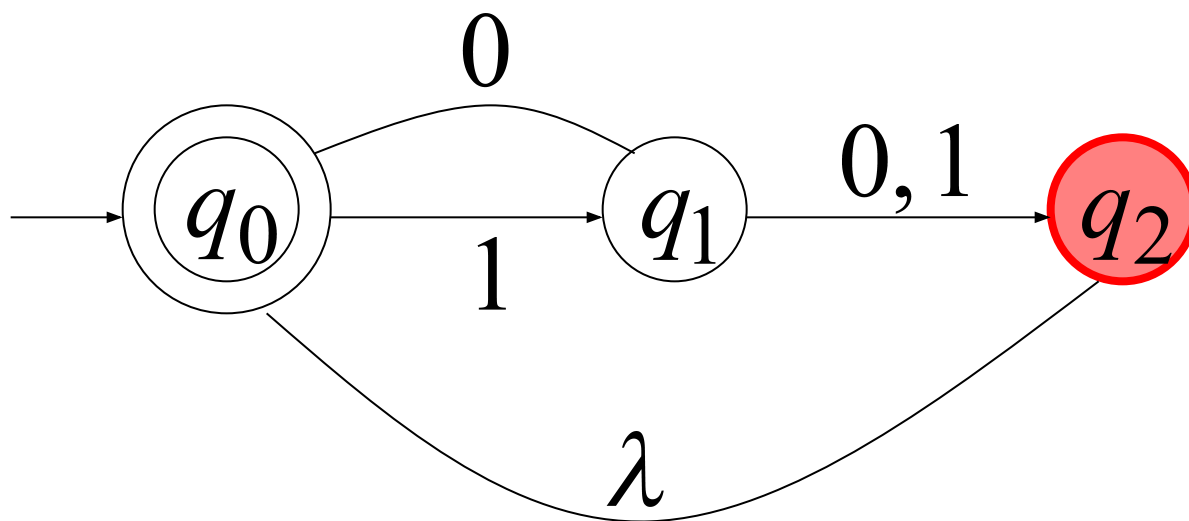
$$\delta(q_1, 0) = \{q_0, q_2\}$$



$$\delta(q_0, \lambda) = \{q_0, q_2\}$$



$$\delta(q_2, 1) = \emptyset$$



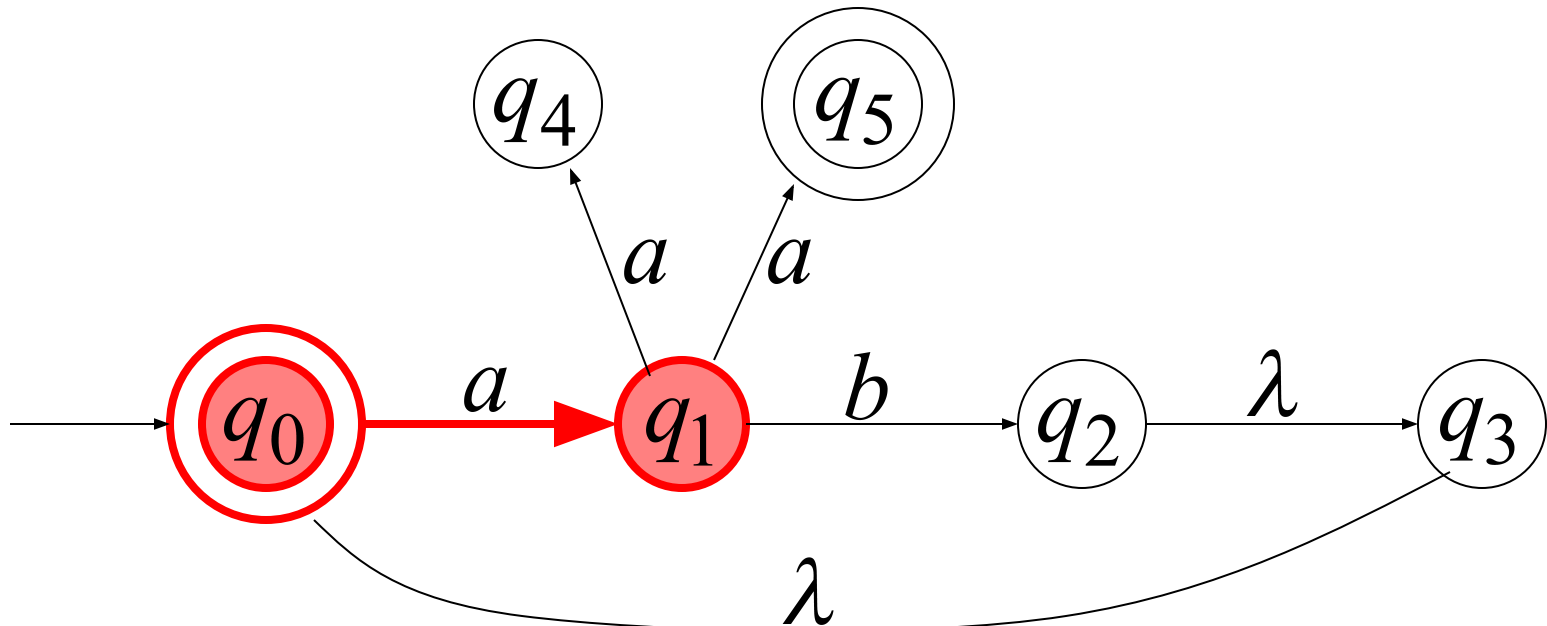
Activity Time



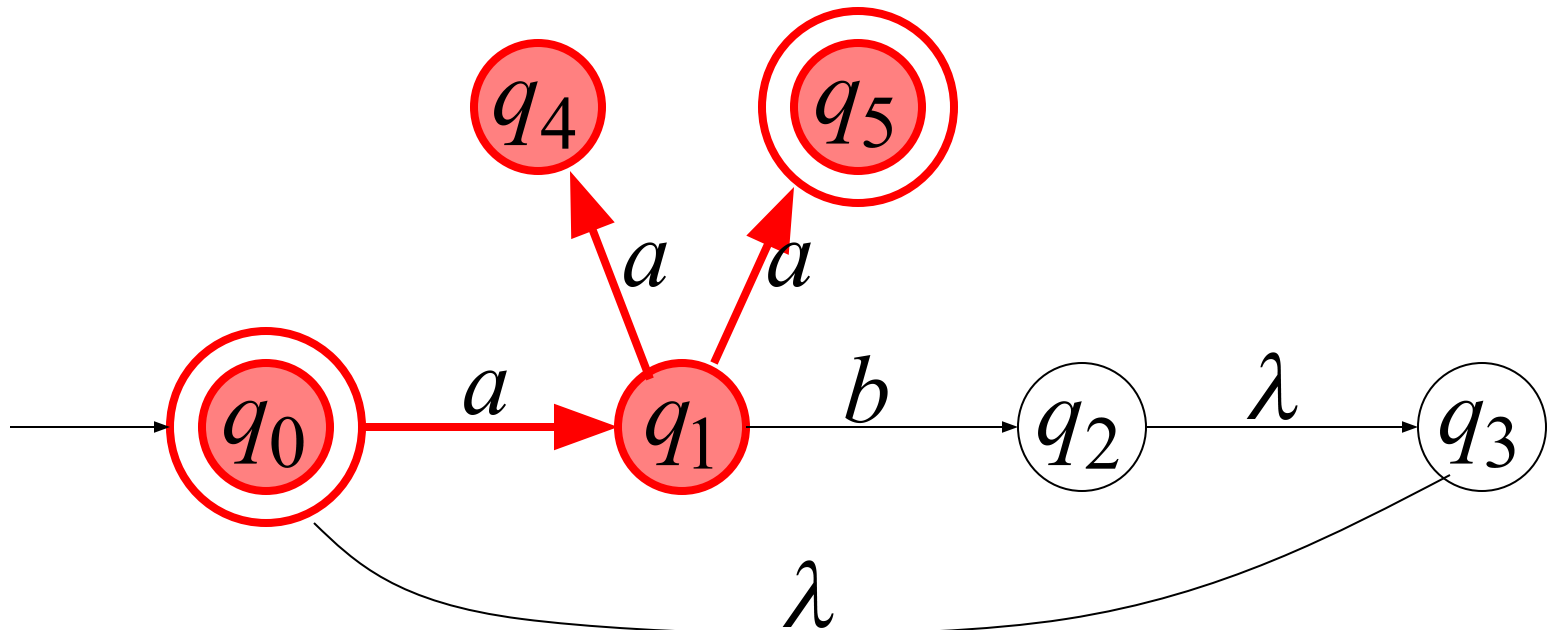
Is the Transition Function for NFA infinite?
Define its Domain X Range..

Extended Transition Function δ^*

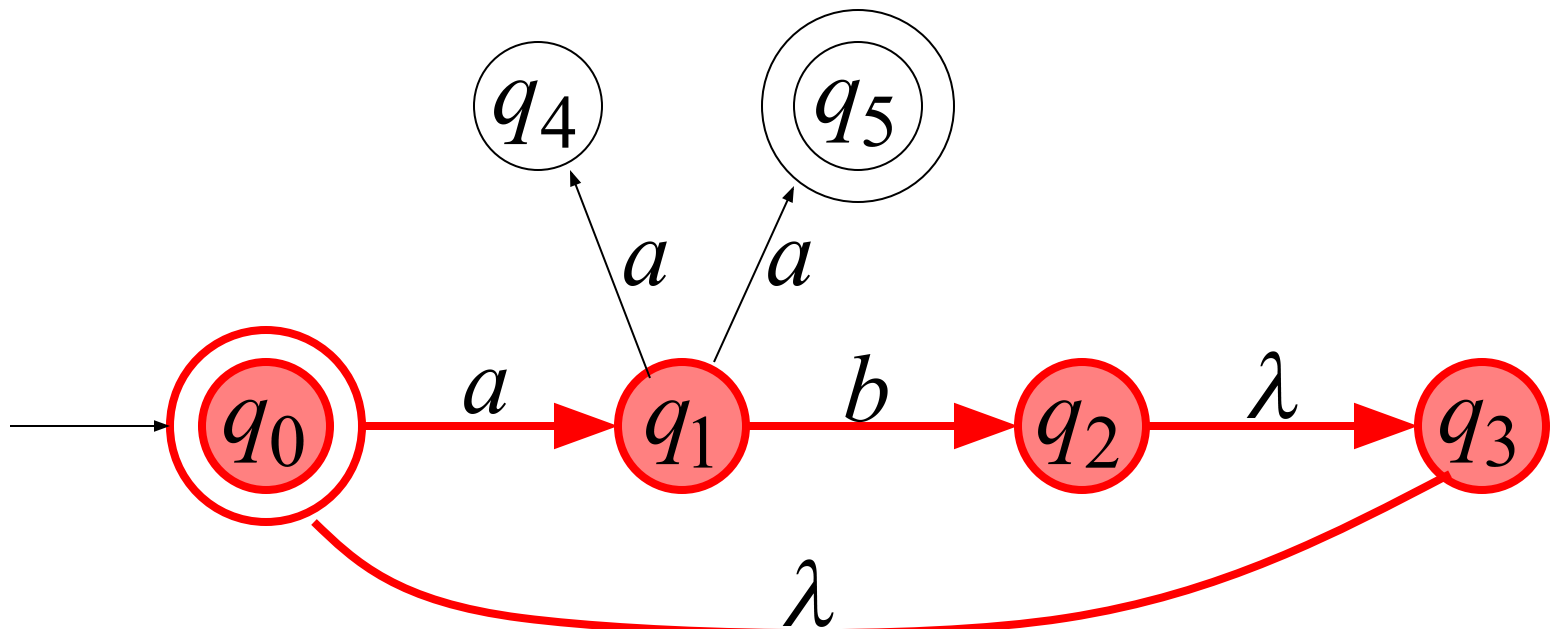
$$\delta^*(q_0, a) = \{q_1\}$$



$$\delta^*(q_0, aa) = \{q_4, q_5\}$$



$$\delta^*(q_0, ab) = \{q_2, q_3, q_0\}$$



Special case:

for any state q

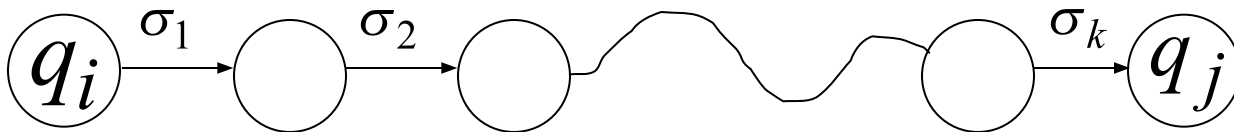
$$q \in \delta^*(q, \lambda)$$

Formally

$q_j \in \delta^*(q_i, w)$: there is a walk from q_i to q_j with label w

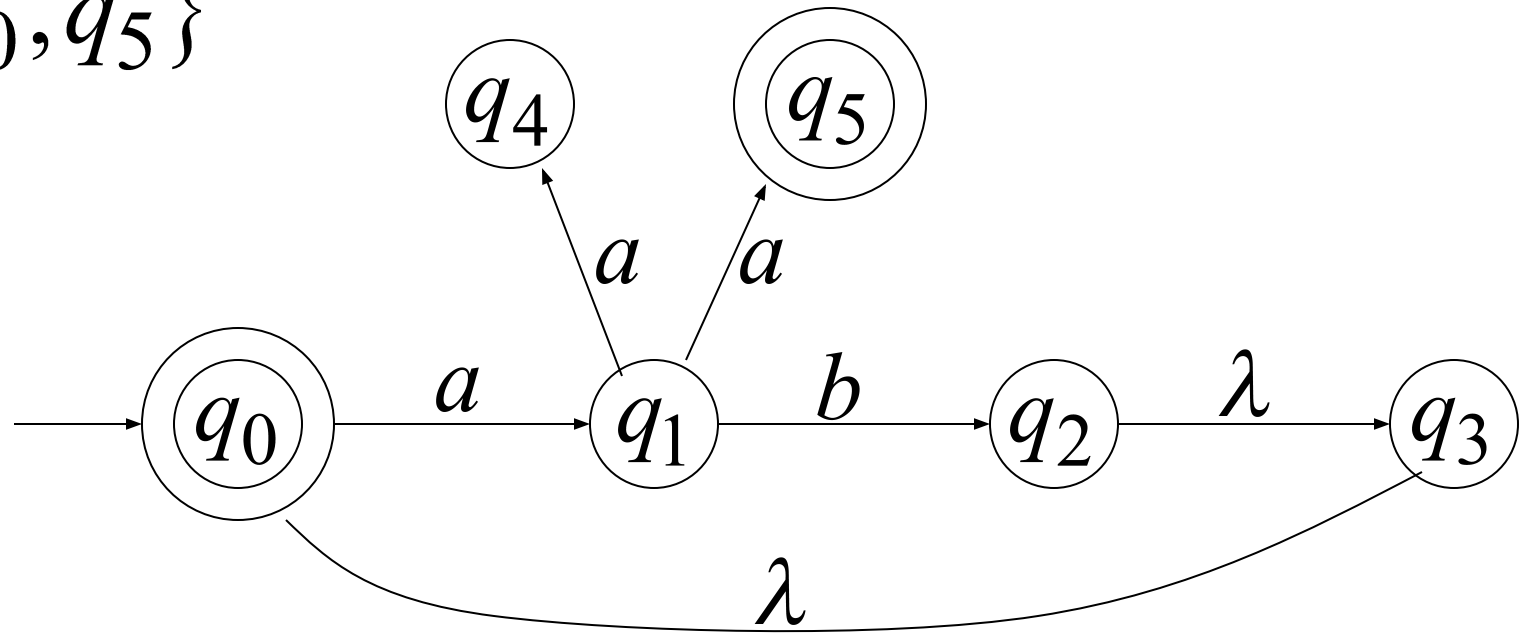


$$w = \sigma_1 \sigma_2 \boxtimes \sigma_k$$



The Language of an NFA M

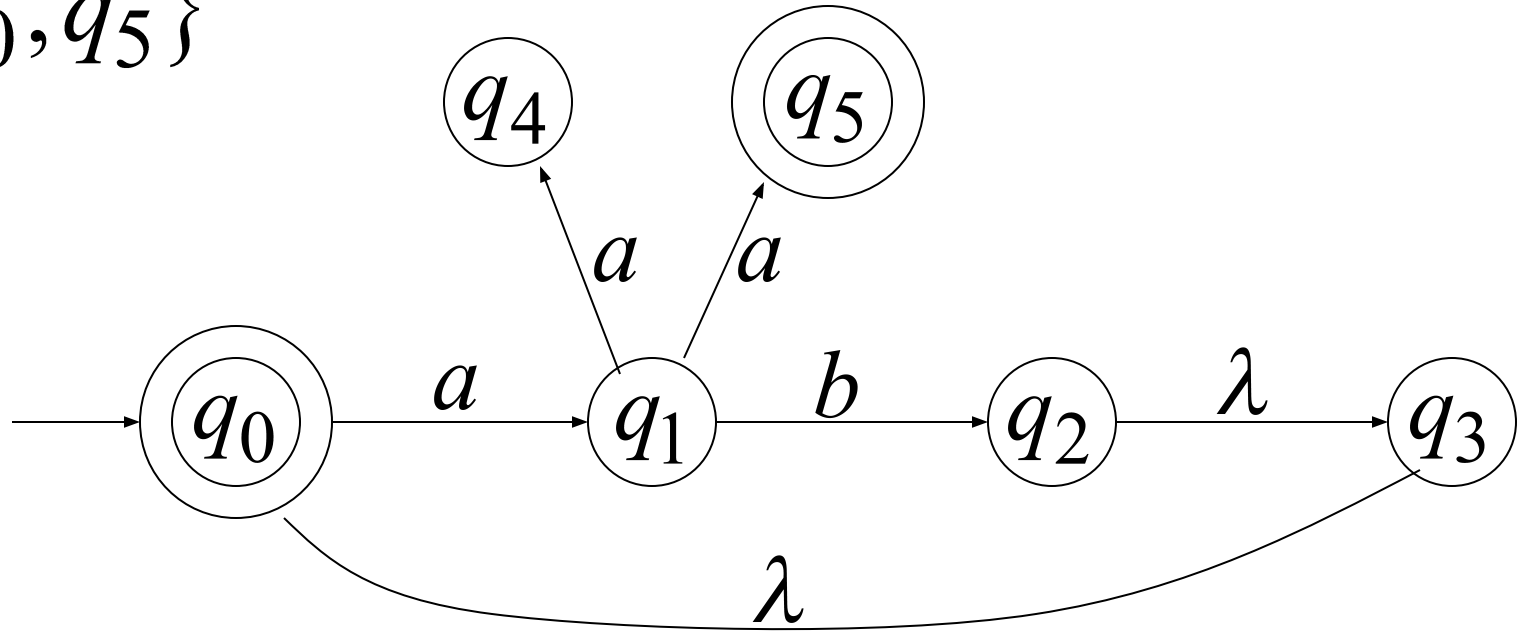
$$F = \{q_0, q_5\}$$



$$\delta^*(q_0, aa) = \{q_4, \underline{q_5}\} \quad aa \in L(M)$$

$\searrow \in F$

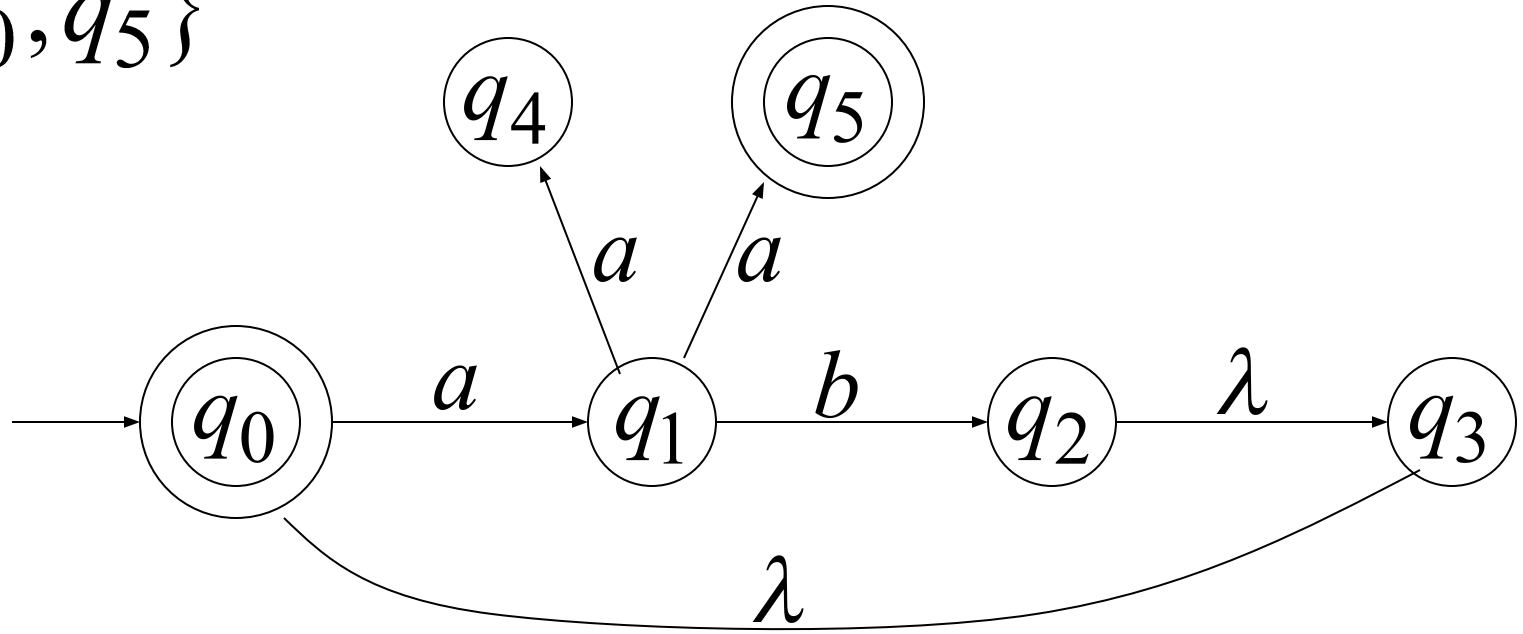
$$F = \{q_0, q_5\}$$



$$\delta^*(q_0, ab) = \{q_2, q_3, \underline{q_0}\} \quad ab \in L(M)$$

$\swarrow$
 $\in F$

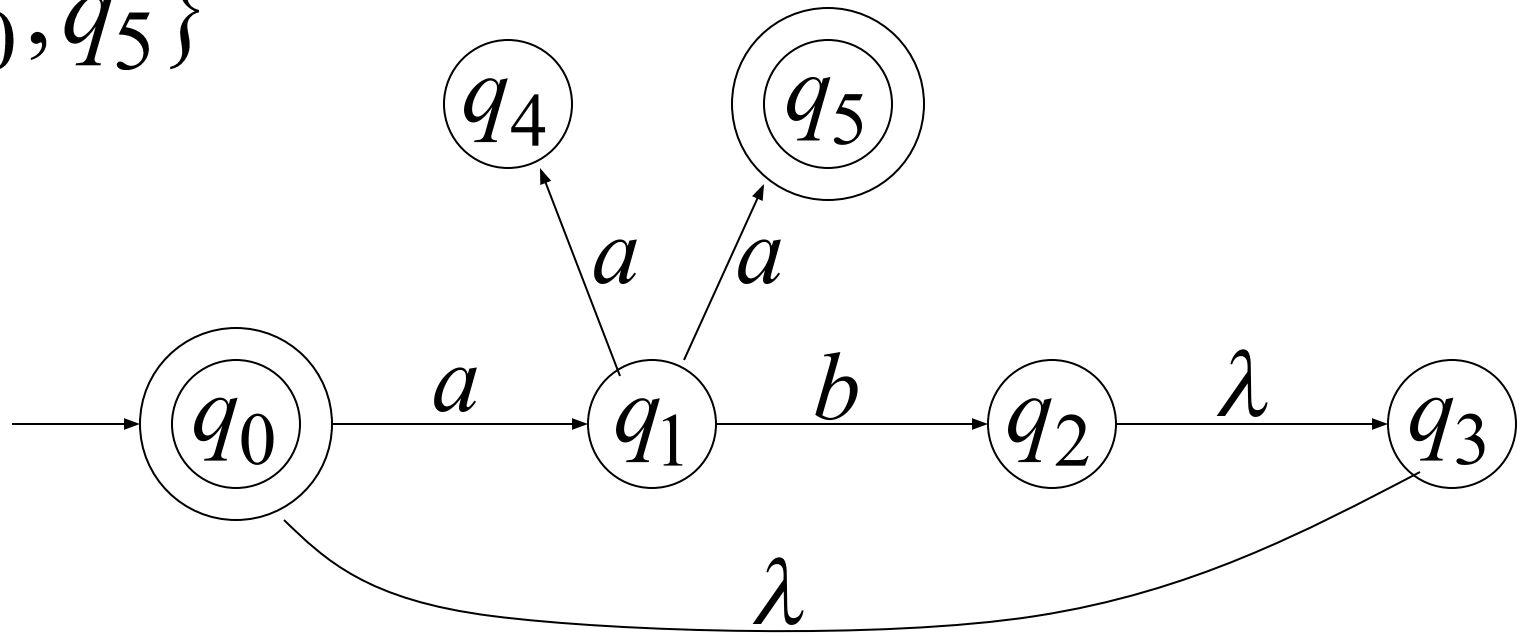
$$F = \{q_0, q_5\}$$



$$\delta^*(q_0, abaa) = \{q_4, \underline{q_5}\} \quad aaba \in L(M)$$

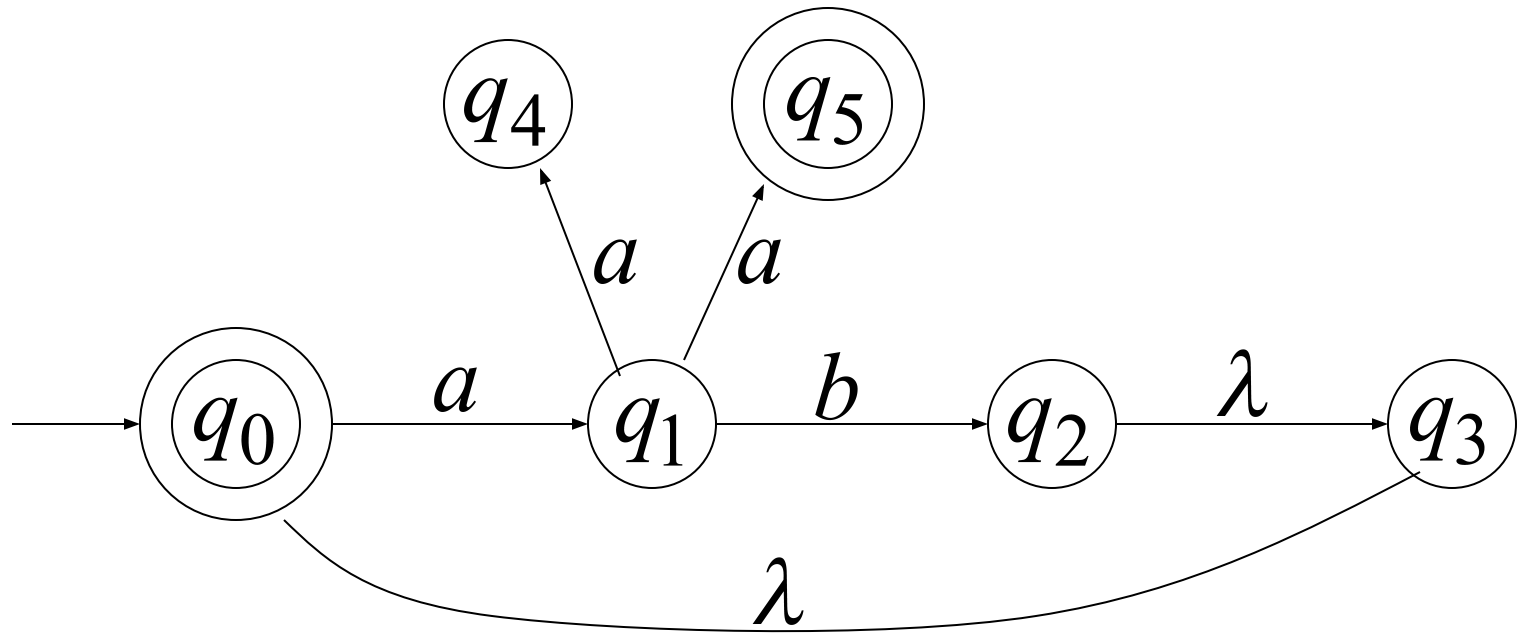
$\swarrow$
 $\in F$

$$F = \{q_0, q_5\}$$



$$\delta^*(q_0, aba) = \{q_1\} \quad aba \notin L(M)$$

$\searrow \notin F$



$$L(M) = \{ab\}^* \cup \{ab\}^* \{aa\}$$

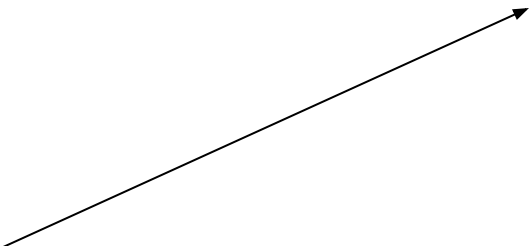
Formally

The language accepted by NFA M is:

$$L(M) = \{w_1, w_2, w_3, \dots\}$$

where $\delta^*(q_0, w_m) = \{q_i, q_j, \dots, q_k, \boxtimes\}$

and there is some $q_k \in F$ (final state)



$$w \in L(M)$$

$$\delta^*(q_0, w)$$

